Research dossier

Report: What Matters in RFQ to Quote

Complete dossier: the main report plus 5 appendices

Documents in this dossier

Main Report	Report: What Matters in RFQ to Quote
Appendix 1	Voice of User: Open Web
Appendix 2	Voice of User: Reddit
Appendix 3	Voice of User: Other Sources
Appendix 4	Domain Rules: Specs, Gauges and Hidden Requirements
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Main Report: Report: What Matters in RFQ to Quote

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Governing question

Section 1

What are the real problems and value props in going from an industrial RFQ to a sent quote, and which matter most to quoting reps and buyers? Within that: what part of the job hurts most and is not already solved by existing tools, so the RudCo demo can win on uniqueness and trust?

It supports two decisions:

1. What the product puts first.
2. Which angle Ken builds into the demo today.

Why now

Section 2

Hacking ends at 4:30 PM ET today, Saturday, October 10, 2026. A video demo is due with the Devpost submission. Uniqueness is one of four judging criteria, each scored out of 5. Trust and explainability is another, and it is the event's theme. The angle has to be picked while there is still time to build it.

Methods

Section 3

- Web research on industry forums, trade press, vendor case studies, and public government RFQs (open-web track).
- A Reddit voice-of-user loop: 25 search pages and 15 threads, saved by hand, read by an agent, every quote checked by script against the saved text (Reddit track).
- Job postings, software review sites, podcast notes, Hacker News, and employee reviews (other-sources track).
- A domain-rules review of gauge tables, ASTM specs, and certification rules, with a 25-line test set (domain-rules track). Thickness gaps come from the track's script, `domain_rules_gauge_calc.py`.
- A comparison of about 15 existing quoting tools from their public pages, plus a survey of public metal price sources (tools and prices track).
- No separate data-munging pass. The only structured data is the Reddit corpus, and the Reddit agent already counted it by script into `voc_records.jsonl`.

Executive summary

Section 4

The job an industrial supplier's quoting rep does all day is simple to say: send the buyer an accurate, competitive quote before a competitor does. The hard part is reading the request. A line like "304 SS sheet 16ga 4x8" packs material, grade, thickness, size and quantity into a code. Each piece has a trap. A gauge number means a different thickness for every metal: 16 gauge is 0.0598 in for mild steel and 0.0625 in for stainless. Two specs written with a slash, like "A513/A500," are two different products. And a requirement the buyer never wrote down, like a domestic-melt rule on a defense order, can turn an ordinary-looking quote into metal nobody can use.

The evidence is clear about where the money goes. The costliest mistakes are the ones that travel: the wrong grade, a missed note, or a certification gap found after the metal has been cut or shipped. Reddit posters describe a $250k job that cost about $450k after the wrong stainless was bought, and a $15-20k profit that shrank to $2-3k because a material price in the system was stale. Speed matters too, but every number claiming that the first quote wins comes from companies that sell quoting software.

The market has already noticed the basic idea. At least four funded startups (Avent, Hexa, Mercura, Proton) say they turn messy RFQs into draft quotes for a rep to approve. One already claims to learn each customer's shorthand. What none of them claims on its public pages is the part the judges care about most: showing where each answer came from, flagging requirements the buyer did not spell out, and asking the buyer a precise question instead of guessing.

That is the opening. The winning demo is a quote that shows its work. Every field is linked to the RFQ words it came from and the table it used. Each line lists its assumptions and its compliance flags, with the trigger text highlighted. When it cannot tell, it asks the buyer one specific question. RudCo's own example makes the case on stage: the two lines its challenge calls "the same item" are 4.2% apart in thickness, and a tool that catches that, and says why, is the trust story in one line.

Researcher's position

Section 5

Build **the quote that shows its work, with hidden-requirement flags as its sharpest feature**. Treat these as one angle, not two. Every line carries three things next to it:

- the RFQ words each field came from, and the gauge table or rule used
- an assumptions and exceptions list: finish not stated, dual cert accepted, spec edition
- compliance flags, each with the words that triggered it highlighted ("DFARS", "SA-240", "domestic", a defense ship-to)

Unclear lines become one-click buyer questions. Lead the demo with RudCo's own example: show that "16ga" stainless and "0.060 in" are not the same thickness, and that the tool asks instead of guessing. Put "learns the buyer's shorthand" last. A vendor already claims it, and it is the hardest to show honestly in one afternoon.

For pricing, show the benchmark's name and date on every line. Say the price tracks published metal cost plus RudCo's margin, never that RudCo is cheapest. The stale-price story is the reason: a price with no visible source and date is how a quote loses money silently.

Confidence: **moderate** (**about 70%**) that this beats the other angles on uniqueness and trust. That the gap in existing tools is real rests on vendor pages, not product tests. The pain evidence comes mostly from job shops, fabricators and buyers, not

RudCo-style distributor reps, who barely appear in public. Confidence is **high** that leading with reading accuracy and the gauge example is right, because judges test against real cases and these cases are checkable.

Claude

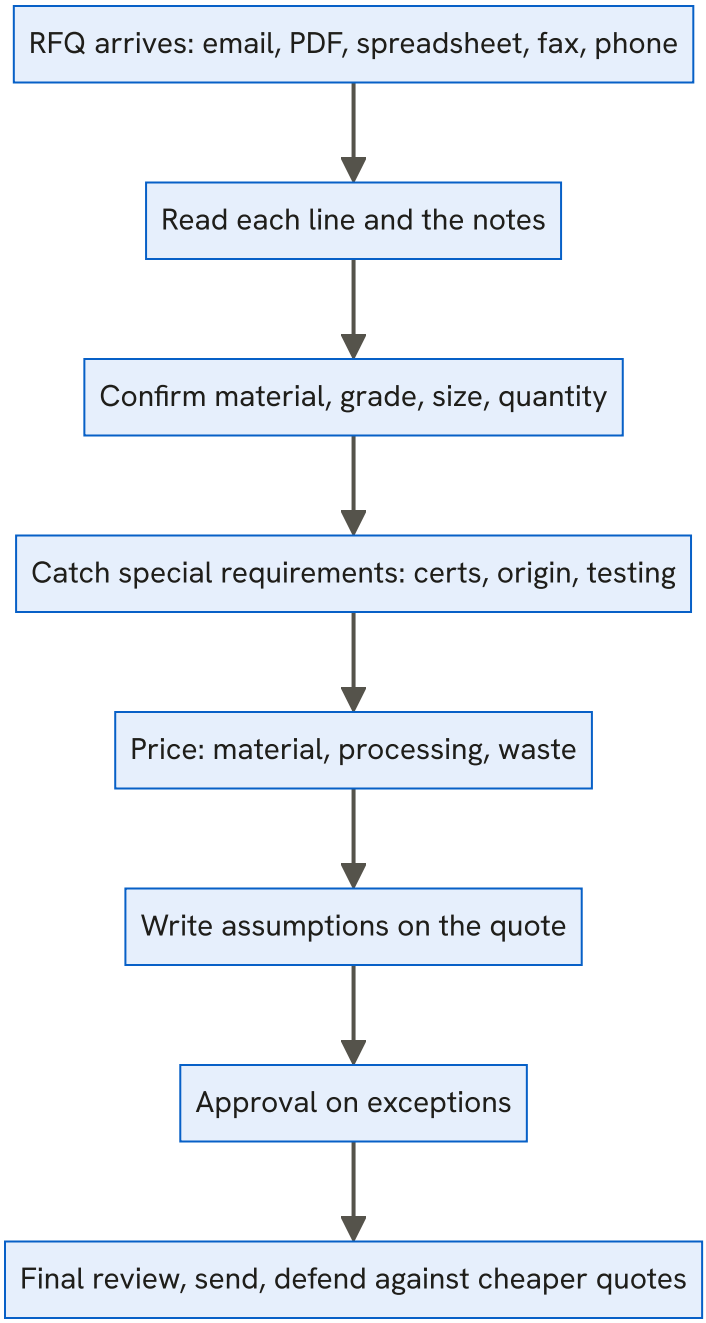
Findings

Section 6

A caveat that applies to everything below. Quoting reps at metals service centers and distributors, the people closest to RudCo's user, almost never post in public. Their voice reaches this report mostly through employers' job postings and vendors' case studies. The first-person voices are job-shop owners, fabricators, machinists, and buyers. Every statistic about speed winning orders comes from a company that sells quoting software. Read the value-prop ranking with that in mind.

6.1 The job, step by step

The rep reads the request, checks every detail, catches special requirements, prices it, writes down the assumptions, and reviews the quote before it goes out. The clearest description is a Toledo steel supplier's estimator posting, read directly. Its thirteen duties run from "Review customer requests, emails, prints, and specifications" through "Identify special requirements such as testing, inspection, tolerances, or origin restrictions" to "Document assumptions, lead times, and any changes clearly on the quote so the customer is never surprised" and "Run a final review of every quote" (Precision Steel Services posting). A Detroit aluminum supplier tells its rep to "Always ask questions, never guess when something is unclear" and to catch "conflicting cert requirements" (Nox Metals posting).



Reddit posters describe the same steps in pieces, plus two that postings leave out: waiting on a boss's approval, and defending against a cheaper quote that "hasn't accounted for X, Y or Z" (r/metalworking).

↳ Drill down

Job postings in full (see appendix)

Steps from Reddit (see appendix)

Steps from vendor case studies (see appendix)

6.2 Where the time goes

Time goes to gathering and checking data, and to re-quoting when a buyer changes a gauge or grade. A Distribution Strategy Group executive writes that on teams he built, people spent 75% of their time on data preparation, and in another passage that they spent 80% of their time hunting through systems, validating data, and assembling quotes by hand. Those are one practitioner's observations, not research (Distribution Strategy Group). Utah Steel says a large job took about two days with three people on quotes, and that when customers asked for quote options in different materials or gauges, it "would require an entire trigger of changes in the quote" (vendor case study, SecturaFAB). On Reddit, one supplier-side poster says repeat RFQs eat "sometimes 10+ hours a week" (r/procurement). The vendor that built Ryerson's quote tool says Ryerson gets a quote request by email every 4.5 minutes, and that quotes "often required multiple clarifying communications" (Concurrency, vendor writing).

Quoting depends on a few senior people. Brunswick Steel says quoting was limited to a select few (SecturaFAB, vendor). At a pipe distributor, newer reps "had to lean on senior folks just to figure out substitutes or where to look" (Canals case study, vendor).

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Time evidence (see appendix)

Ryerson and Puget Sound (see appendix)

6.3 The value props, ranked

Accuracy comes first, then speed, then a complete, comparable quote, then rep time saved. The ranking weighs how much each costs when it fails and who is saying it.

Rank	Value prop	Strongest evidence	Who says it
1	**Avoid costly errors**	Wrong stainless bought: "a $250k job that ended up costing close to $450k" (r/metalworking). 1,800 pieces of tubing cut and machined, then left unusable, after 303 arrived instead of 304 (Evident, sells analyzers)	Suppliers, self-reported
2	**Speed to the first full quote**	A buyer post scored 222 on Reddit compares shops that never reply with tools that quote "within 24hours" (r/CNC). Win-rate claims all come from vendors	Buyers; vendors
3	**A complete, comparable quote**	Buyers' own RFQs demand unit price, delivery date, shipping included, and SAM registration with FAR and DFARS reps and certs (USACE RFQ). Buyers complain that suppliers "technically answer the RFQ but don't answer it on the same basis" (r/procurement)	Buyers
4	**Rep time on repeat work**	"10+ hours a week" (above). A pipe distributor's inside sales manager says in a vendor case study he has typed some items "tens of thousands of times—and still made mistakes" (Canals)	Suppliers

Speed numbers are vendor numbers. A 2020 Paperless Parts survey says 67% of buyers expect a quote in under 24 hours (Paperless Parts). A consultant repeats that 50% of buyers pick the first responder, citing unnamed technology providers. A CloudForge customer says win rate rose over 50% (Modern Metals). Each source sells quoting software or was chosen by one. No poster on Reddit ties faster quoting to a measured win rate.

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Costly errors on Reddit (see appendix)

Value props from vendors (see appendix)

Value props from employers (see appendix)

6.4 The costliest mistakes

The mistakes that cost the most are the ones found after the metal is cut or shipped: wrong origin, wrong grade or spec family, and missed notes. The domain-rules track ranks errors by whether they can be fixed after shipping:

1. Wrong or missing origin (DFARS, Buy America). Metal melted abroad cannot be repaired.
2. Wrong grade or spec family: 304 vs 316, A36 vs A572-50, A500 vs A513, A53 vs A500.
3. A missing certification. Fixable only if the heat number was tracked.
4. Wrong gauge table. It is silent, because the quote still looks normal.
5. Wrong unit or position, such as feet vs inches or a trailing "x4".
6. Sheet vs plate at the 3/16 in boundary.

That ranking is judgment from consequences. No public source prices a misread RFQ line. Reddit adds the categories rules miss: notes on the print. "Early in my career I missed a note calling out weld size" (r/metalworking). "Missed the part where it said the welds are to be purged" (r/metalworking). It also adds stale prices: material pricing "wasn't getting updated like I thought it was" (r/metalworking). Most of the Reddit error stories come from one thread, with 13 commenters.

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Error ranking (see appendix)

Error stories (see appendix)

Wrong-grade cases (see appendix)

6.5 Messy lines and the gauge trap

RudCo's own example of "the same item" is two different items. In the challenge, "304 SS sheet 16ga 4x8" and "Stainless 304, 0.060 in, 48 x 96" stand for one item. They are not. 16 gauge stainless is 0.0625 in. 0.060 in matches 16 gauge mild steel, 0.0598 in. The gap is 4.2% of thickness (Metal Supermarkets gauge chart; gap from the domain-rules script).

Gauge	Mild steel	Galvanized	Stainless	Aluminum
14	0.0747	0.0785	0.07812	0.06408
16	0.0598	0.0635	0.0625	0.05082
18	0.0478	0.0516	0.050	0.04030

Reading the wrong table puts stainless about 4.5% thick and aluminum up to 24.2% thin (at 10 gauge). Tube wall has its own table again: 16 gauge tube wall is 0.065 in (Wikipedia, Sheet metal).

Real public RFQs are messy in ways a parser can test. An Air Force RFQ writes "STAINLESS Steel 2B T-304 - 16 Gage x 4 x 8 or similar" with no units, under lines that use feet (BidNet, FA302024Q0014). A 2021 State of Iowa RFQ gives a short name of "Stainless Tubing 1 x 1 x .05" over a description saying ".065 wall" (Iowa RFQ, 2021). A 2018 Iowa RFQ writes "Square Tube A513-A500 Grade B," which names two different products (Iowa RFQ, 2018). The tracks together give the builders 25 test lines with readings and questions, plus 19 verbatim lines from Reddit.

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Gauge tables and the 25-line test set (see appendix)

Government RFQ lines (see appendix)

Reddit lines (see appendix)

6.6 Hidden requirements

Employers expect reps to catch requirements the buyer may not spell out, and buyers say the paperwork is what they pay for. "Those certs are what you're paying for," says a machinist with a top-voted comment, score 107 (r/Machinists). A Navy RFQ for HY-80 plate makes full traceability, mill test reports, and DFARS 252.225-7009 compliance conditions of the quote itself (HigherGov, N4215826QE068).

Most triggers sit outside the line, so the tool has to read the whole RFQ and stay quiet when nothing triggers. The grade decides whether a rule can apply. Stainless counts as a specialty metal under DFARS because of its chromium (DFARS 252.225-7009). The contract, the ship-to, or a priority rating on the PO decides whether it does apply. The 2018 Iowa RFQ, 49 lines long, names no certification or origin rule at all. A tool that flags everything would fail the product brief's own test against needless questions.

The safe flags are the ones written in the line. These words imply paperwork directly: `SA-240` (ASME code material), `A500` (ships with a mill test report), `304/304L` (dual cert), `NACE`, `DFARS`, `Buy America`, "domestic", `3.1`, `PMI`.

"DFARS" and "domestic" get blurred. One buyer asks for "DFARS Compliant 60 series aluminum. Also has to be made in the USA" (r/Machinists). A commenter there says aluminum is exempt from DFARS. That claim is unchecked. By the clause's chemistry definition, it depends on the alloy, so the tool should ask rather than rule.

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What triggers each requirement (see appendix)

Cert stories (see appendix)

Requirements in job postings (see appendix)

6.7 What existing tools already do

The basic loop is already sold, so "AI reads the RFQ and drafts a quote" will not win on uniqueness. Avent, Hexa, Mercura, and Proton each say they turn messy RFQs into draft quotes with a rep approving (Avent, Hexa, Mercura, Proton). None of them is metals-specific at the line level. No public product reads gauge, grade, and ASTM spec for a US metals supplier.

Angle	Claimed by a vendor?	Gap
Quote shows its work	No vendor links each field to the RFQ words. Nearest: confidence scores (Cassidy)	Open
One-click buyer questions	One vendor lists a Message Center to centralize communication as a planned update (DigiFabster)	Open
Learns buyer shorthand	Proton claims it learns "your customers' nicknames." Canals claims it learns each customer's needs	Taken
Catches hidden requirements	No vendor mentions MTRs, DFARS, Buy America, or implied certs	Most open

These are claims read from marketing pages. No tool was tested, so "open" means not claimed publicly.

Users distrust AI output they cannot check. "As anything with AI it is highly risky and hit-and-miss, as all models hallucinate AF you still have to check manually so there is no point" (r/procurement). A reviewer praises Paperless Parts for "identifying features of a part that we might overlook," and complains its estimated cycle time is "a little off at times" (SaaSworthy, G2 reviews). Trust follows whether the user can see why.

↳ Drill down

Tool comparison table (see appendix)

User voice on tools (see appendix)

6.8 Price explanations the data can honestly support

Public data gives market cost, not competitors' prices, so the quote should say its price tracks published metal cost plus margin. The best free sources are the BLS producer price indexes for steel mill products and aluminum mill shapes on FRED (steel, aluminum), and the IMF monthly nickel and aluminum prices on FRED (nickel). Nickel drives stainless alloy surcharges. North American Stainless posts a monthly fuel surcharge publicly (NAS). Fastmarkets, CRU, and Platts are what the industry actually references, and they are paid. The stale-price story in 6.4 is the reason to show each benchmark's name and date on every line.

↳ Drill down

Price sources table (see appendix)

6.9 The four angles, scored

Show its work and hidden requirements score highest. Buyer questions are a strong supporting feature. Shorthand learning scores lowest.

Angle	User pain	Gap in existing tools	Shows trust to judges	Verdict
Quote shows its work	Employers require documented assumptions and a final review. Buyers want an exceptions column next to every line	Open	Highest: judges can check any field	**Build**
Catches hidden requirements	Certs are what buyers pay for. Missed notes and wrong grade are the costliest errors	Most open	High, if each flag shows its trigger text	**Build, inside show-its-work**
One-click buyer questions	"Always ask questions, never guess." A buyer-side commenter says to ask for "the exact missing field"	Open	High: "asks instead of guessing" is the brief	**Build as the fallback**
Learns buyer shorthand	Real, with thin evidence	Taken (Pro-ton, Canals)	Low in a one-day demo	**Skip**

A fifth idea comes out of the Reddit track: an exceptions and assumptions list beside each line. It is folded into "shows its work" above rather than treated as its own angle.

↳ Drill down

Angle analysis, Reddit (see appendix)

Angle analysis, other sources (see appendix)

Angle analysis, domain rules (see appendix)

Citations

Section 7

The challenge

- Get Hired Hack, Devpost

Job steps and value props

- Precision Steel Services estimator posting
- Nox Metals SDR posting
- Distribution Strategy Group, Hopkins, July 2025
- SecturaFAB, Utah Steel
- SecturaFAB, Brunswick Steel
- Concurrency, Ryerson
- Canals, Puget Sound Pipe & Supply
- Paperless Parts, responding too slowly
- Modern Metals, CloudForge

Reddit threads

- r/procurement, repetitive RFQs
- r/procurement, comparable quote fields
- r/procurement, comparing supplier quotes
- r/metalworking, things missed on a fab quote
- r/CNC, US machine shops
- r/Machinists, military fasteners
- r/Machinists, DFARS aluminum

RFQs and domain rules

- Metal Supermarkets gauge chart
- Wikipedia, Sheet metal
- USAF RFQ FA302024Q0014, BidNet
- USACE steel RFQ, BidNet
- State of Iowa RFQ 245, 2018
- State of Iowa RFQ 245, 2021
- Navy HY-80 RFQ, HigherGov
- DFARS 252.225-7009
- Evident, the case of the cracked weld

Tools and prices

- Avent, YC launch
- Hexa, YC launch
- Mercura, YC launch
- Proton.ai, order and quote automation
- Cassidy RFQ agents
- DigiFabster AI Quote Agent, 3D Printing Industry
- SaaSworthy, Paperless Parts reviews
- FRED, PPI steel mill products
- FRED, PPI aluminum mill shapes
- FRED, IMF nickel price
- North American Stainless news

Appendix 1: Voice of User: Open Web

Read this first

Track: voice of user, open web (no Reddit). Run 2026-10-10. Covers the job steps (Question 0), pain in practitioners' own words (Question 1), and the costliest reading errors (Question 3). Messy line examples (Question 2) are in their own section near the end.

The answer in one paragraph. The open web gives a clear picture of what hurts, but very little of it is a metals service-center rep talking in their own words. Most of the "voice" is vendors (quoting software sellers) describing their customers' pain, plus government RFQs that show what buyers actually ask for. The strongest, most consistent signals are:

Speed matters, but the numbers behind "first responder wins" all come from sellers of quoting software.

Picking the wrong item from many look-alike candidates is the named cause of wrong shipments at a steel bar distributor.

A change of gauge, grade, or alternate material "would require an entire trigger of changes in the quote" (Utah Steel case study).

Quoting depends on a few senior people.

Buyers' own RFQs ask for unit price, delivery date, shipping included, and proof paperwork (mill test report, certificate of compliance, origin). A quote that skips any of these does not answer the RFQ.

Gaps, stated up front.

I found no first-person posts from metals service center inside sales reps. Searches for them returned job postings and vendor pages.

LinkedIn posts were not reachable through search. Zero LinkedIn posts are cited.

Buyer-side annoyance with quotes is thin. One 2008 forum thread and one 2020 vendor survey are all I read.

Practical Machinist and Eng-Tips users are machine shops and engineers, not service-center reps. They show the quoting mindset, not the service-center workflow.

Metal Center News and Modern Distribution Management (MDM) pages mostly returned blocked or paywalled text. I read one MCN piece (ERP software) in full and one Industrial Distribution piece. I did not find an MCN article on quoting that I could read.

No steel service center publishes a hit rate or quote turnaround benchmark that I could find.

How to read the evidence labels in this file.

Read directly means I read the page text in a browser or a local file. Quotation marks around words from these pages are exact.

Read through a summarizing fetch tool means a tool returned a summary of the page. I paraphrase these and do not put the words in quotation marks.

Search summary only means I saw the result only in a search snippet. These are leads, and I say so each time.

A claim from a company that sells quoting software is marked **vendor claim**.

Contents

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Cleanup needed

Question 0: The job steps, who does them, where time goes, what a better quote wins

Section 1

The steps, as the sources describe them

No source lays out a full step list for a metals service center. Pieced together from the sources below, the work runs in this order:

1. **Intake.** The RFQ arrives by email, PDF, spreadsheet, fax, or phone. A steel bar distributor's inside sales team handles about 200 RFQs and purchase orders a week (eAlliance case study, vendor case study, read through a summarizing fetch tool).
2. **Decode the line.** Work out which item the buyer means. At that same distributor, a part has many attributes, so a search returns 20 to 30 candidates and the rep has to judge which is right (same eAlliance source, vendor claim).
3. **Check stock and availability.** In stock, in production planning, or a substitute (same eAlliance source).
4. **Price it.** Estimators measure parts, track down material prices, and gather estimates for machine run time, cleaning, painting, labor, and shipping (SecturaFAB case study of Brunswick Steel, vendor case study, summarizing fetch tool).
5. **Get approval on exceptions.** In distribution generally, the rep checks prices across separate systems, requests an exception, and waits for finance or pricing

to approve, often by email (Conga author, Distribution Strategy Group, June 2026, summarizing fetch tool; the author works for Conga, which sells quote software).

6. **Send the quote and chase it.** Errors found later send the quote back through approvals (same Conga source).

Who does each step

- **Quoting sits with a few experienced people.** Brunswick Steel (a Manitoba service center) says quoting was limited to a select few, and its old process leaned on Excel (SecturaFAB case study above, vendor; the speaker is Dallas McInnes, sales manager, via summarizing tool).
- **Product selection depends on staff knowledge.** The Conga author says finding the right SKU in a large catalog often relies on experienced staff's knowledge instead of standard process.
- **Even in machine shops, one estimator is the bottleneck.** A Practical Machinist poster who was buying a company wrote: "I need to improve the way we respond to quotes- only have 1 estimator and response time is too slow, I believe it is losing us business." Source: user Neeman, Dec 21, 2020, Practical Machinist thread (read directly).

Where the time goes

- **Large jobs at a steel supplier.** Utah Steel (Cedar City, Utah) said a large job took about two days to quote with three people on quotes. Requests for alternative materials or gauges forced a full round of manual changes. After buying SecturaFAB, it reports about two hours. Vendor case study, so these are the vendor's published numbers (Utah Steel case study, summarizing fetch tool).
- **Data work, not selling.** Brian Hopkins, COO of the consulting firm Distribution Strategy Group, writes that on teams he built, people spent 75% of their time on data preparation, and in another passage 80% of their time hunting through systems, validating data, and assembling quotes by hand. He says complex quotes sometimes took his team up to eight hours. It is one practitioner's observation, not research (Hopkins, July 2025, summarizing fetch tool).
- **Retyping.** A pipe, valve, and fitting vendor (SalesAPE, which sells RFQ software) says sales engineers waste hours retyping customer bill-of-material spreadsheets into the ERP and that spec parsing takes days instead of minutes. Its numbers are illustrative, not measured (SalesAPE, July 2026, vendor, summarizing fetch tool).

- **Machine shops.** Paperless Parts (a quoting software seller) says it can take up to two hours to quote one part. This is from a sponsored article in Modern Machine Shop (MMS, Nov 2021, sponsored by Paperless Parts, vendor, summarizing fetch tool).

What a faster or better quote wins, and who says so

Value prop	Who says it	Source and caveat
Orders won by speed	Quoting software sellers and one consultant	Paperless Parts survey: 67% of buyers expect a quote in under 24 hours (up from 53% in 2019). Paperless Parts blog, vendor survey. Sample size not stated on the page I read. A different search snippet attributed a 2020 survey of 400+ buyers, and a later piece says 68% from 370 buyers. The two numbers are not the same survey.
Orders won by speed	Hopkins	More than 40% of potential wins lost when quotes take over 24 hours, and 50% of buyers pick the first responder. He cites unnamed "leading technology providers." Unverified (Hopkins).
Orders won by speed	Prokeep	Distributors with faster, more accurate quotes are 35 to 50 percent more likely to win. This is Prokeep's own data. It sells order automation. Search summary only. Prokeep
Speed matters to the seller	Jaden Seely, Utah Steel	He said "Speed is everything in the sales world" (summarizing tool; vendor case study, Utah Steel).
Fewer wrong shipments	eAlliance, steel bar distributor	Wrong part selections led to wrong shipments, which the case study calls costly. No numbers given. Vendor case study.
Fewer returns	SalesAPE	Misreading specs such as NACE MR0175 sour service or pressure classes leads to inaccurate pricing and costly job-site returns. Vendor marketing, no numbers.
Margin protection	Hopkins	A misplaced decimal, a forgotten freight charge, or outdated pricing can turn a profitable quote into a loss. Opinion from the same consultant.
Less dependence on experts	Brunswick Steel; Conga author	The software "reduces a highly skilled task to a task that others can confidently take on" (McInnes via summarizing tool; vendor case study).
More time to sell	eAlliance; Hopkins	Staff freed from clerical work. Vendor and consultant claims.
More precise pricing	Seely, Utah Steel	He said estimates went "from guesstimating to exacting" (summarizing tool; vendor case study).

A counterweight worth knowing. Not every RFQ is worth a fast quote. A search snippet from an MDM piece (What to Fix Before Investing in CRM, MDM) says one rep responds to every RFQ and spends hours on projects with little chance of winning, and that qualifying earlier matters more (search summary only; the page would not load). A machine-shop owner says the same thing in his own words: "the overall objective is still to put in the minimum unpaid work until you actually get the gig." Source: user implmex, Apr 18, 2019, Practical Machinist thread (read directly).

Win-rate benchmarks. I found no metals-specific hit rate. A generic quoting glossary from a software vendor gives 25-40% for established customers and 10-20% for new prospects (Zigaflow glossary, search summary only, vendor, not metals).

Question 1: Pain in practitioners' own words, and buyer annoyances

Section 2

What practitioners actually say (read directly)

All quotes below are from pages I read in a browser.

Quoting fast without over-working it.

- User DanASM, Apr 2, 2020, Practical Machinist: methods to expedite quoting: "It takes me less than 5 min to get a material quote from 4 major suppliers with competitive pricing and delivery." Also: "I have a few customers who want quotes within 72 hours." And: "You cant waste time quoting every job down to the second."
- Same user on how he builds a number: after requesting quotes 4-5 times "I have a reliable number to quote my customers from experience." This shows the buyer of metal (a shop) gets quick supplier quotes. The service center is the one answering those four-supplier requests.

Buyers shop around by sending the RFQ to everyone.

- User MikeHalloran, Sep 12, 2011, Eng-Tips: steel plate: "Either way, just fax a RFQ describing what you want to all the service centers you can find, and pick from the quotes." He adds that "Steel service centers are happy to supply cut rectangles to exact dimensions, but charge a little extra for the cross-cuts and for handling the 'small' pieces." The reader of an RFQ is one of several, and extras like cross-cuts change the price.

Small errors multiply on volume parts.

- User implmex (a shop owner), Apr 18, 2019, Practical Machinist: "with volume parts a small error has big cumulative consequences." This is a machine shop, but the same math applies to a wrong unit or grade on a large order.

Customers ask for odd things, and the quote has to handle them.

- User wheelieking71, Oct 10, 2014, Practical Machinist: "I do also get asked A LOT, "how much if I supply material?"" Buyers also ask a supplier to quote against left-over pieces that may not exist.

Software and ERP for service centers (the supplier side, read directly).

- Paul Parsons, VP of sales and marketing at 4GL (an ERP seller), in Metal Center News, Feb 27, 2026 (Service Center Software): "There are things that happen in metal distribution centers that don't happen in generic distribution companies. Metal service centers are cutting into plate. They have remnants that need to be taken into consideration. All of those things generic software struggles with."
- Shawne O'Connor, director of Paragon Consulting Services (maker of METALWARE): "Being able to work with multiple units of measures, mixing and matching your units. Those are key factors in managing your inventory." Both are ERP sellers, so they have a reason to say generic software falls short. For our decision, the useful point is that multiple units of measure and remnants are everyday realities for service centers.

Pain stated by vendors and consultants (labeled, read through a summarizing tool)

- **Handoffs.** The Conga author says every handoff between teams adds a chance of miscommunication and each manual step adds risk, and that sales pushes for speed and lower prices while finance enforces margin limits, which creates exceptions and slow approvals.
- **Alternates.** Utah Steel says requests for alternative materials or gauges force a full round of manual changes. This is the closest thing to a rep-side pain about changes after the first quote.
- **Wrong candidate chosen.** eAlliance: 20 to 30 look-alike candidates per part, rep judgment decides, wrong choice means a wrong shipment.

- **Vague requests.** A fabricator, Staub Manufacturing, writes: "nothing slows down the quotation process more than having to stop and ask questions." Source: Staub, six things you need for fast accurate fabrication quotes (summarizing tool; Staub is a fabricator and the page doubles as marketing). It lists what RFQs often lack: material type, nominal thickness, grade, special sourcing rules, finish, packaging, quality requirements, volume, and deadline.
- **Questions back to the buyer.** This supports the "one-click buyer questions" idea in a general way: stopping to ask is the slow step, so questions that are cheap for the buyer to answer matter. It does not prove buyers dislike being asked.

Buyer annoyances (thin)

Evidence that buyers dislike receiving quotes is weak. What I found:

- **No response to a quote request.** A buyer, user silentrunning, Mar 18, 2008, wrote that he requests quotes from new vendors and never hears back, and that lead times for basic raw materials and hardware seem unreasonable (Elsmar Cove thread, summarizing tool). It is from 2008 and about general sourcing.
- **Suppliers' view of buyers.** Same thread (summarizing tool): suppliers produce many full quotes and land few, dislike "price fishing" with three-quote requests when the mind is already made up, and note that buyers ask for a firm quote when unsure they can buy soon. This is the supplier side of the same relationship.
- **Speed expectation.** The Paperless Parts survey above (vendor survey).
- **What the buyer's RFQ asks for, which is the best evidence of what a quote must contain.** Real government RFQs (see the next section) ask for unit and total price, a delivery date estimate, shipping included in the price, and proof paperwork. A USACE RFQ says "DO NOT quote shipping separately" and says offers are judged on price, speed of delivery, and past performance. A buyer who gets a quote that leaves these out has to ask again.
- **Quote validity and price firmness.** I could not find buyer complaints about this. Supplier terms pages show metals quotes valid for as little as one business day for volatile metals, and others price at the "prevailing price" at time of shipment (Avalon, search summary only; not a metals service center). Treat as background for the price-benchmark track.

Question 3: Costliest reading errors

Section 3

The evidence is a mix of forum stories and vendor marketing. No source gave a cost per error for a service center.

1. Wrong grade or material variant

- **A stainless tubing mix-up (Evident Scientific, which sells XRF analyzers, so it has a reason to tell this story).** A manufacturer ordered SS304 tubing and received SS303, which cracks when welded. The paperwork said 304. The error showed up in non-destructive testing after 1,800 pieces were cut and machined, so those pieces were unusable. The article says incorrect certificates and markings raise the risk at the mill, service center, and fabricator. (Evident: The Case of the Cracked Weld, summarizing tool, vendor.)
- **A 410 versus 410-S shipment.** A forum thread (Engineers Edge) describes a brake rotor maker who ordered 410 stainless, got 410-S, found out at the first heat-treat step, and was told the vendor would replace the material but not pay the roughly $7,200 in process cost. A reply argues that once you accept the shipment it is your material. I could not open the page (bot check), so this is **search summary only**: Engineers Edge thread.
- **Takeaway.** When the wrong grade is discovered downstream, the buyer's processing cost can far exceed the material cost, and who pays is a fight. This is the strongest case for "ask, don't guess" on grade.

2. Wrong thickness from a gauge number

- The same gauge number means different thicknesses by metal. For 16 gauge: steel 0.0598 in, galvanized 0.0635 in, stainless 0.0625 in, aluminum 0.0508 in. Source: Xometry, "The Weird Thing About Gauge" (summarizing tool; Xometry sells manufacturing services). Xometry says the table is not comprehensive.
- A gauge-to-thickness mismatch can mean a part that is out of tolerance. I found no source that put a dollar cost on a gauge error, so the cost is inferred, not documented.

3. Wrong part picked from look-alike candidates

- eAlliance (steel bar distributor): wrong selections led to wrong shipments. No numbers. Vendor case study.

4. Missing or late paperwork

- **Navy HY-80 plate RFQ.** The buyer requires full traceability, complete physical and chemical mill test reports showing MIL-S-16216 compliance on delivery, DFARS 252.225-7009 specialty-metals compliance, and a DX-A1 rated order. The quote itself must confirm MTR capability and include a certificate of compliance. A quote that misses these is non-responsive. Source: HigherGov listing of RFQ N4215826QE068 (summarizing tool). This is a hidden-requirements example in plain sight: the paperwork is in the RFQ, but a rep who reads only the item line misses it.
- **Certs arriving after the material.** On an Elsmar Cove quality forum thread, users say it is common for material to arrive before certs, and certs can get lost when a distributor splits a lot. Summarizing tool: Elsmar thread, page 2.
- **A buyer rejecting material for a missing cover letter.** User LittleInch, Dec 11, 2019, on Eng-Tips: ASME material certificate? (read directly), answered a poster whose client wanted a certificate tied to an ASME edition: "I find it difficult to believe someone would "reject" materials on these grounds." That shows buyers do set paperwork conditions that surprise suppliers.
- **What a certificate has to show.** User marcolino21576, Jan 24, 2012, on Eng-Tips: standard for material test reports (read directly): a 3.1 certificate "must reflect property of the material you are purchasing." So the type of certificate changes what the supplier must deliver.
- **Cost of a missing MTR.** One traceability software vendor page claims a 14-day hold and a $240K recall cost per event, with no source (OxMaint, search summary only). I do not rely on it.

5. Wrong unit

- I found no documented unit-error story. The MCN piece above (read directly) shows service centers mix units of measure routinely, which is the setup for the error.

What this means for the demo (inference, not from a source)

- The most damaging errors are the ones that travel: grade and certification gaps that surface only after material has been processed or inspected. Those are the places where "ask, don't guess" saves the most.
- The gauge and unit errors are common and easy to demonstrate, but no source says they cost the most.

Messy line examples

Section 4

Public solicitations only. I removed the names and contact details of government staff. "Verbatim" means I read the text directly.

A. USAF consumable metals RFQ (verbatim, read directly)

Source: BidNet listing of FA302024Q0014 (Air Force, Gulfport MS, closed 2024-02-01). The government's SAM.gov notice is here.

```
(80 Each) Steel Plate (Low Carbon), A36 3/8" x 4' x 8' or similar
(40 Each) Steel Plate (Low Carbon), A1011- 11 Gage x 4' x 8' or
similar
(100 Each) HR Galvanized Sheet 26 Gage x 4' x 10' or similar
(20 Each) Aluminum Plate, 5052-32 - .063 x 4' x 8' or similar.
(20 Each) STAINLESS Steel 2B T-304 - 16 Gage x 4 x 8 or similar
```

Also in the notice: "Please include Shipping/Delivery in quote. Quested delivery within 30 days ARO. Forklift on site, need to package all steel in stacks of 10 max, due to lift capability."

Why it is messy:

- "or similar" on every line.
- The last line writes "4 x 8" with no units; the lines above use feet.
- "T-304" and "2B" run together.

- "Aluminum Plate, 5052-32 - .063" is 0.063 in, which is thin enough to be sheet, not plate (my reading, check with the domain track).
- "A1011" is the ASTM spec for hot-rolled sheet, but the line calls it "Steel Plate" with an 11 gage thickness.
- The 16 gage stainless should read as 0.0625 in per the Xometry table above, not 0.060 in.

B. USACE steel RFQ (verbatim, read directly)

Source: BidNet listing, USACE St. Paul District, Pleasant Valley, Iowa, April 2023.

```
ASTM A-572 G50 1.25" x 96" x 144" 3 EA
ASTM A-36 0.75" x 8" x 20' 10 EA
ASTM A-36 W12 x 79 Beam 20' 3 EA
ASTM A-36 Angle 0.75" x 8" x 6" x 20' 4 EA
```

Why it is messy: mixed inches and feet on one line, "G50" for Grade 50, the quantity at the end of the line, and "Angle 0.75 x 8 x 6" without saying which leg is which. The notice also says to give "unit pricing for each item," include a delivery date estimate, and ship FOB destination with shipping not quoted separately.

C. State of Iowa RFQ, 2021 (verbatim, read directly from the PDF the fetch tool saved)

Source: State of Iowa RFQ 245, DB22245005, dated 2021-07-16. Each line has a short name and a longer description. They do not always agree.

```
Line 6   Stainless Tubing 1 x 1 x .05
         Stainless Steel Tubing Type 304, 1" x 1" x .065 wall
thickness x 20' Long
Line 8   Tubing 1 x 1 x .065
         SquareTubing A513/A500 Grade B, 1 x 1" x .065 wall thickness
x 24' Long
Line 12  Cold Rolled Sheets
         Cold Rolled Sheets CQ/A366, 48" x 96" x 18 Gauge
Line 13  Hot Rolled Sheets
         Hot Rolled Sheets CQ/A569, 48" x 96" x 10 Gauge
Line 14  Stainless Sheets  (filed under "Steel, Mild: Bars, Plates,
Rods, Sheets, Strips, Tubes, etc.")
         Stainless Steel Sheets Type 304-2B, 60" x 120" x 14 Gauge
Line 2   HR Angle 1 x 1 x 1/8
         Stainless Steel Angle Type 304, 1" x 1" x 1/8" x 20' Long
Line 11  Tubing 1 1/2 x 1 x .083
         Rectangular Tubing A513/A500 Grade B, 1 1/2" x 1" x .083 wall
thickness x 24' Long
```

Why it is messy, from the text itself:

- **Line 6**: the short name says ".05" and the description says ".065". The two disagree.
- **Line 2**: the short name says "HR Angle" (hot rolled) and the description says "Stainless Steel Angle Type 304". The short name and description point to different materials.
- **Line 14**: stainless sheet filed under the "Steel, Mild" commodity heading.
- **Lines 5, 8, 10, 11**: "A513/A500 Grade B" lists two specs with a slash. In my background knowledge, A513 is mechanical tubing and A500 is structural tubing, so the slash hides a real choice. Verify with the domain track.
- **Lines 12, 13**: "CQ/A366" and "CQ/A569" cite specs that, as I recall, were later replaced by A1008 and A1011. The Air Force RFQ above uses "A1011". Verify with the domain track.
- **Line 8**: "1 x 1" x .065" has an inch mark on the second number only.
- The state RFQ also says a bid is valid for 90 days.

D. Navy HY-80 plate (read through a summarizing fetch tool)

Source: HigherGov listing of N4215826QE068. The summary reported: "PLATE, ALLOY STEEL, GRADE HY-80", spec MIL-S-16216, 5 ft x 20 ft x 0.625 in (5/8") thick, quantity 1, an NSN, and an equivalent reference "(or equal)". Requirements: full traceability, complete physical and chemical MTRs, DFARS 252.225-7009, priority rating DX-A1. The thickness is given as 5/8 in and 0.625 in in two forms. The paperwork requirements are the point of this one.

E. Other leads (search summary only; check before use)

- A Coast Guard RFQ 70Z04026Q60204Y00 is reported to include "3/8 A-36 HR PLT 96 X 240" (5 each), a 1/2" A-36 plate in the same size (15 each), a 1" plate at 96 x 240, and a 3/4" plate at 48 x 96, plus a 14-inch A992 wide-flange beam. I could not open the notice. One summary page says delivery by Oct 7, 2026 and another says a close date of Aug 19, 2026, which conflict.
- An Air Force Sheppard AFB RFQ reportedly lists "PL3848 ST Plate 3/8 X 4' X8' A36" (fifty). I did not open it.
- A Zoro retail listing URL for a stainless sheet ends with "3044-16gx48x96" (product slug with grade, gauge, and size run together). I read the URL only: Zoro. Useful as a shorthand example, not an RFQ.
- Vendor page SalesAPE lists "316L stainless steel seamless pipe" and "Duplex 2205 vs. 304/304L stainless" as examples of ambiguous metallurgy (summarizing tool, vendor).

Same item, different words (built from the sources above)

All four describe 16 gauge stainless 304 sheet, 4 x 8 ft. Each comes from a different real RFQ or listing style:

- `STAINLESS Steel 2B T-304 - 16 Gage x 4 x 8 or similar` (Air Force RFQ, verbatim)
- `Stainless Steel Sheets Type 304-2B, 60" x 120" x 14 Gauge` (Iowa RFQ, different size, same style)
- `3044-16gx48x96` (retailer slug)
- Per the Xometry table, the thickness these should resolve to is 0.0625 in for 16 gauge stainless.

Search terms and sources tried

Section 5

Terms that returned useful material:

- "quoting RFQ turnaround time service center inside sales quote response time hit rate"
- "Practical Machinist estimating RFQ quoting pain points"
- "Modern Distribution Management quote speed win rate distributors quoting process inside sales"
- "Metal Center News quoting automation service center"
- "steel service center AI quoting RFQ email PDF extraction inside sales"
- "sheet metal gauge confusion wrong thickness ordered stainless galvanized"
- "RFQ 304 stainless sheet 16 ga 4x8" and "RFQ steel plate A36 mill test"
- Domain-limited searches on practicalmachinist.com, eng-tips.com, and metalcenternews.com

Terms and sources that came up empty:

- First-person posts from service center inside sales reps (only job postings and vendor blogs).
- LinkedIn posts on quoting from metals reps. The search tool did not return any LinkedIn pages.
- Metals-specific hit rate or quote-to-order benchmarks.
- MSCI (Metals Service Center Institute) public material on quoting. Not found.
- Any dollar cost for a wrong-gauge or wrong-unit error.
- A DFARS rejection cost for a distributor.

Access problems. Plain automated fetches returned 403 for Practical Machinist, Eng-Tips, MDM, MCN, BidNet, and Engineers Edge. I read several of those pages through a normal browser tab. The MDM articles returned only a headline or a different page when loaded, and the Engineers Edge page showed a bot check. I did not touch reddit.com.

What I would not trust yet

Section 6

- All "first responder wins" statistics. Their sources are software sellers or unnamed.
- Any "X% faster" figure. They are vendor self-reports with no baseline.
- The Engineers Edge $7,200 story and the Coast Guard line items until someone opens the originals.
- The statement that ASTM A366 and A569 were replaced. It is my recall, flagged for the domain track.

Cleanup needed

Section 7

Nothing to delete. I opened one browser tab while reading pages and closed it. Scratch files from this run are in the session scratchpad only (`.../scratchpad/ow/`, including blocked-page HTML from failed curl attempts) and are not in the repo.

Appendix 2: Voice of User: Reddit

Contents

Coverage gaps (read first)

Short version

Findings

What this says about the four candidate angles

Failure mode notes

Corpus coverage

Track: Voice of user, Reddit (research questions 0 to 4). Written 2026-10-10 for the RFQ sharper-angle brief. Every quote below is copied from a saved Reddit payload and checked against it by script. Each carries a subreddit, month, score, and link. Typos are the posters' own.

Coverage gaps (read first)

Section 1

- **No quoting rep at a service center or distributor appears in this corpus.** The supplier voices are job-shop owners, fabricators, machinists, and one sales rep at a plastics job shop. RudCo's own role is missing. The closest matches are the anonymous poster in the repeat-RFQ thread and the plastics rep in the "who builds your quotes" thread. The open-web track has to cover distributor reps.
- **Reddit gives no measured time-per-step.** The only number is one poster's "10+ hours a week". Where time goes is inferred from what people complain about, and that is labeled as inference below.
- **The costly-error evidence comes from one thread.** The thread "What's the dumbest little thing you've seen get missed on a fab quote" (Aug 2026) holds 19 of the 94 quotes in this file, from 13 commenters. These are unverified self-reports. Dollar figures are the posters' own.
- **Round 1 search pages were mostly noise.** 24 of 25 search pages returned results; item 18 (r/purchasing) returned zero. Most hits were hobbyists, construction take-

off estimators (r/estimators), and unrelated viral threads. r/engineering, r/MechanicalEngineering, r/manufacturing, and r/Welding each gave two or fewer relevant threads.

- **Tool builders posted several of the buyer-side threads.** The flag file lists 13 vendor or promotional items across 4 of the 15 threads (7 of them in the repeat-RFQ thread). Posts by builders are not used as evidence. Their comment sections are, with the builder's account marked where it appears.
- **Reddit's saved files are missing some comments.** The CNC thread reports 256 comments and the file holds 182. The procurement quote-format thread reports 24 and holds 15. See the coverage manifest for every thread.
- **Dates run Aug 2023 to Aug 2026.** Seven of the 15 threads are from 2026. Nothing here predates 2023.
- **Unchecked technical claims.** A commenter says aluminum is exempt from DFARS. One account, score 2, no source. The domain-rules track should check it.
- **Demographic skew.** Reddit skews technical and early-adopter. The people here chose to post a question or a story. People who quietly lose a quote do not.

Short version

Section 2

- **The most-named pain is the cost of a mistake that slips into a quote, then the amount of manual re-work per RFQ.** Missed notes, wrong material, and stale prices produce the biggest numbers ($250k job costing about $450k; $15-20k expected profit shrinking to $2-3k). Repeat manual RFQ handling produces the most-repeated complaint (10+ hours a week).
- **Hidden requirements are real, but the evidence comes from four threads of machinists, a welder, and one startup builder, not from distributors.** The most vivid line: the same bolt can come with certs or without them.
- **Buyers say speed and a visible reason matter.** A 225-score post compares US shops that never reply with online tools that quote in 24 hours and give design feedback.
- **Of the four candidate angles, the evidence supports "the quote shows its work" and "catches hidden requirements" most. "Learns the buyer's shorthand" has the thinnest support.** Details in the last section.

Findings

Section 3

RQ0: The job steps, who does them, where time goes

Claim 0.1. Reps describe the steps as: a request arrives by email, someone digs up past prices and suppliers, someone prices it, a boss approves it, and the rep chases the boss. Reddit gives the sequence only in pieces; this table is assembled from several posters, so it is a reconstruction.

Step	Who (as posters describe it)	Evidence
1. Request arrives by email	Whoever monitors the inbox	r/procurement, Apr 2025, link
2. Read the print, spec, and notes; find what is missing	Estimator or rep; sometimes the owner	r/Machinists, Dec 2024, link, r/metalworking, Aug 2026, link
3. Look up past prices, material cost, and suppliers	Same person	r/procurement, Apr 2025, link, r/metalworking, Aug 2026, link
4. Price material, labor, outside work, hardware	Estimator, GM, or (at one past employer) temps with a price book	r/sales, Feb 2026, link, r/sales, Feb 2026, link, r/metalworking, Aug 2026, link
5. Approval	GM or owner; margin rules decide who can skip it	r/sales, Feb 2026, link, r/sales, Feb 2026, link
6. Send, then defend against a cheaper quote	Rep or owner	r/metalworking, Aug 2026, link, r/metalworking, Nov 2025, link

Claim 0.2. Manual handling of repeat RFQs is the most direct time complaint, and it comes from the supplier side. (6 records in 2 threads; 13 records in 4 threads across the manual-work themes.)

- “we constantly get quote requests via email, and it turns into a mess of checking inboxes, digging up previous prices, googling suppliers, and replying manually” (r/procurement, Apr 2025, score 22) link

- “It ends up eating way more time than it should (sometimes 10+ hours a week)” (r/procurement, Apr 2025, score 22) link
- “Most exhausting part of the process is feeling like you’re starting from scratch with each RFQ.” (r/procurement, Apr 2025, score 5) link
- “quotes were done by an army of untrained temps” (r/sales, Feb 2026, score 3) link
- “manually calculate based on pricing guidebooks, physical printouts where you'd match the feature to a price and add it all together with a calculator” (r/sales, Feb 2026, score 1) link

Counter-evidence: one commenter says "Googling?" does not match their organization, so the pain varies by size.

- “I think we may have different sized organisations” (r/procurement, Apr 2025, score 14) link

Another commenter, who says they manage a large indirect-spend budget, describes the status quo as spreadsheets and email:

- “90% of our RFPs are excel and email.” (r/procurement, Apr 2025, score 3) link

Claim 0.3. Approval is a bottleneck the rep cannot control, and the rep carries the blame for it. Single thread (a plastics job shop, not metals), so this is an anecdote with a plausible fit to distributors.

- “I have to prove I'm not at fault by showing when I emailed the RFQ package to him.” (r/sales, Feb 2026, score 2) link
- “a delay to the customer is my fault, regardless of reason” (r/sales, Feb 2026, score 2) link
- “the expectation is that I follow up with him daily on projects we need to quote” (r/sales, Feb 2026, score 2) link

One reply shows how margin rules can remove the approval step, and another says reviewing is easier than writing from scratch:

- “sales people have authority to quote anything with a Contribution Margin (Labor and Materials ONLY cost) of greater than X%” (r/sales, Feb 2026, score 2) link
- “It’s easier for someone to review then to do.” (r/sales, Feb 2026, score 11) link

Claim 0.4. The value props people name themselves, in order of how much weight the evidence carries. This ranking is mine, built from the counts below. It is a judgment call, not a poll.

Value prop	What posters say	Threads	Source side
Avoiding costly errors and lost margin	Missed notes, wrong material, stale material prices turn into five- and six-figure losses	15 records in 1 thread	Suppliers
Speed to a first response	Buyers pick tools that answer in 24 hours with design feedback; shops admit ghosting	4 records in 1 thread	Buyers and suppliers
Rep time saved on repeat RFQs	"10+ hours a week"	6 records in 2 threads	Suppliers
Comparable, explained quotes	Buyers want the same basis across quotes and the reason behind each number	11 records in 4 threads	Buyers
Compliance paperwork done right	Certs and traceability are what the buyer pays for	16 records in 4 threads	Both

No poster ties faster quoting to a measured win rate. That link is assumed, not shown, in this corpus.

Claim 0.5. Buyers who get no reply describe moving to online tools that answer in a day. The top-scoring buyer complaint is about response, and a shop-side commenter confirms that shops ghost.

- “90% of the time they don’t even ever reach out to you” (r/CNC, Mar 2026, score 222) link
- “send in a quote on an online quoting tool, get quote within 24hours or sometimes instantly. Get dfm analysis that you can view” (r/CNC, Mar 2026, score 222) link
- “China will just build a test batch for trials before a Western company has got back in touch on the RFQ.” (r/CNC, Mar 2026, score 33) link
- “I am guilty of ghosting such clients.” (r/CNC, Mar 2026, score 26) link

Caveat: this is a buyer shopping one-off machined parts from job shops, not a distributor's customer. Shop owners in the same thread also say cold, badly designed requests earn no reply.

- “Stupid tight tolerances for no reason.” (r/CNC, Mar 2026, score 80) link
- “it's not their job to unfuck your design for a quote.” (r/CNC, Mar 2026, score 27) link

RQ1: Pain, in their words

Claim 1.1. Buyers say quotes from different suppliers arrive on different bases, so they cannot be compared. (11 records in 4 threads.)

- “One gives unit price only. Another gives MOQ but no lead time. Another mentions “shipping extra” but doesn’t say how much.” (r/procurement, Jan 2026, score 1) link
- “the biggest issue is when suppliers technically answer the RFQ but don’t answer it on the same basis.” (r/procurement, Aug 2026, score 1) link
- “an EXW and a DDP quote aren't comparable numbers at all” (r/procurement, Aug 2026, score 3) link
- “Spec revision because the silent version is lethal: three suppliers quoting three different revisions look like a price spread but are actually a scope spread” (r/procurement, Aug 2026, score 3) link
- “"quoted per your drawing" with no revision number is the tell” (r/procurement, Aug 2026, score 3) link
- “a minimum comparable-data gate before evaluation: unit price with currency and UOM, lead time definition, MOQ, Incoterms, payment terms, and validity date.” (r/procurement, Feb 2026, score 1) link

Account note: the three procurement threads behind these quotes were started by accounts tied to quote tooling. One links its own tool, one discloses working on supplier and quote tooling, and one was called a vendor by another commenter. The commenters quoted above are not flagged in the saved files. The provenance and exceptions-column quotes later in this file come from the account that started the comparable-fields thread.

Buyers' fix on Reddit is a mandatory template, not a tool:

- “Yes we have a standard RFT response template we expect them to fill in and non-negotiable fields.” (r/procurement, Jan 2026, score 9) link
- “You also don't need to do something so complex where a formatted (standardized) excel sheet can do the job.” (r/procurement, Feb 2026, score 2) link

Claim 1.2. Suppliers and buyers describe incomplete requests that need a question before anyone can price, and the question is often asked informally or skipped. (8 records in 5 threads.)

- “What would you guys do about missing dimensions for angles, lengths radius etc. would you go back to the customer for that info or just go ahead and make them up as it’s all so open?” (r/Machinists, Dec 2024, score 2) link
- “What angle is needed?” (r/sheetmetal, Dec 2024, score 15) link
- “Not a single dimension mentioned” (r/sheetmetal, Dec 2024, score 9) link
- “From my end, it is just difficult to gauge a cold-rfq, but I do try to help out people as much as I can if they put in some notes in the email.” (r/Machinists, Apr 2026, score 3) link
- “instead of asking "can you confirm your quote?" you can ask for the exact missing field that blocks comparability” (r/procurement, Aug 2026, score 1) link

The last quote is from a buyer-side thread, but it states the one-click-question idea in plain words: ask for the exact missing field, not for a general confirmation.

Claim 1.3. Suppliers describe price pressure from a cheaper quote that left things out. Single thread.

- “it turns out the other supplier hasn't accounted for X, Y or Z.” (r/metalworking, Aug 2026, score 4) link

Claim 1.4. Same RFQ, wildly different prices. One post asked for a price on a one-off long turned part, shown as an image. The commenters guess prices from about $1k to $20k. These are commenters guessing, not real quotes, but the spread shows how much depends on assumptions that the RFQ does not state.

- “Ornamental only? 1k all in.” (r/Machinists, Dec 2024, score 2) link
- “$3200 labor only” (r/Machinists, Dec 2024, score 19) link
- “$20k” (r/Machinists, Dec 2024, score -2) link

Claim 1.5. On tools, users say instant online quoting is fast and cheap to try but misses the human feedback and can run high. Other tool mentions in the corpus are single-line.

- “Xometry is near instantaneous and free, and won’t waste another human’s time.” (r/Machinists, Apr 2026, score 9) link
- “I've tried Xometry in the past, but they've been over 2-5x the costs I get back.” (r/Machinists, Apr 2026, score 2) link
- “QuoteFx, QuoteWin are both SaaS that manage quotes and pricing data” (r/supplychain, Dec 2024, score 3) link

Claim 1.6. Commenters distrust AI-structured quote output and vendors fishing for product ideas. One thread, so a signal about trust, not a count.

- “As anything with AI it is highly risky and hit-and-miss, as all models hallucinate AF you still have to check manually so there is no point.” (r/procurement, Jan 2026, score 2) link
- “I spot ChatGPT answer I stop answering.” (r/procurement, Jan 2026, score 6) link
- “Just another vendor fishing to help build their app” (r/procurement, Jan 2026, score 9) link

RQ2: Messy line items (test cases)

Claim 2.1. The same item shows up under several names, and the source often does not resolve which is meant. Each row is copied from the post or comment. "Reading" is given only where the source makes it clear; otherwise it says so. (17 records in 6 threads.)

Line as written	Reading, and where it comes from	Source
“Spec and quote was for 316ss sch80 pipe.”	316 stainless, Schedule 80 pipe. The same story later says the purchased material was "304ss sch40", so the two lines differ in grade and wall.	r/metalworking, Aug 2026, link; r/metalworking, Aug 2026, link
“I ordered $80,000 worth of 7” XS seamless pipe to make 700 caps.”	"XS" is not defined in the thread. In general industry use it means extra-strong wall. That meaning is from outside the source.	r/metalworking, Aug 2026, link
“made from 3/8" 316 stainless plate”	3/8 inch thick, 316 stainless plate.	r/metalworking, Aug 2026, link
“steel sqare pipe witz 5mm thickness”	Square pipe (tube), 5mm wall, as written by a non-native English speaker. The story then substitutes 4mm.	r/metalworking, Aug 2026, link; r/metalworking, Aug 2026, link
“one place wants $685.77 for a 43x67" 1/2" thick plate HRPO plus $150 for plasma cutting it to that size from the stock sheet. Another place which is a huge steel supplier but still deals with the public quoted me $417.24 but it's just A36 and not HRPO.”	The poster treats HRPO and A36 as different products. A commenter's reading is in the next row.	r/metalworking, Nov 2025, link
“As far as I know, HRPO is just acid-dipped hot rolled A36”	This commenter says HRPO is hot rolled A36 that has been acid-dipped to remove mill scale. One commenter's view.	r/metalworking, Nov 2025, link
“301 stainless steel, 26 gauge, brushed finish. 10 units for prototyping, then 100.”	301 stainless, 26 gauge, brushed finish, 10 then 100 units. A commenter equates the gauge to ".018" thick; another warns about "half hardened 301". Temper is a hidden attribute.	r/sheetmetal, Dec 2024, link; r/sheetmetal, Dec 2024, link; r/sheetmetal, Dec 2024, link
“It may be 304 versus 301 stainless for hvac ducts.”	A commenter casually swaps 304 for 301. Whether that is acceptable depends on the spec; the thread does not say.	r/sheetmetal, Dec 2024, link

Line as written	Reading, and where it comes from	Source
“Looking for a supplier for DFARS Compliant 60 series aluminum. Also has to be made in the USA”	The buyer uses "DFARS compliant" and "made in the USA" together. A commenter separates them: domestic is doable, a DFARS statement on the cert is the hard part. The buyer says the customer wants domestic.	r/Machinists, Aug 2023, link; r/Machinists, Aug 2023, link
“I need 3:16 7075 round”	Unclear. Possibly 3-1/16 inch diameter 7075 round bar; the source does not say.	r/Machinists, Aug 2023, link
“Assuming 46" of 4" A36”	A material assumption: 46 inch of 4 inch A36 bar.	r/Machinists, Dec 2024, link
“Mcmaster had a 3ft bar of 1018 at 3-7/8 dia for $388.”	A different assumption for the same drawing: 1018 bar at 3-7/8 inch. Same RFQ, two bar sizes and two grades.	r/Machinists, Dec 2024, link
“Does not say anywhere that this is cylindrical.”	The drawing does not say the part is cylindrical, so a plasma-cut reading is possible.	r/Machinists, Dec 2024, link
“ePlastics natural ULEM 1000, 10ft.”	Likely a typo of Ultem 1000. The source does not say so outright.	r/Machinists, Apr 2026, link
“ULTEM 1000 or Polycarbonate”	Same thread, material named as "ULTEM 1000 or Polycarbonate".	r/Machinists, Apr 2026, link
“McMaster prices for 5ft of Ultem PEI is like $24.75/bar”	A third name, "Ultem PEI", for what looks like the same plastic.	r/Machinists, Apr 2026, link
“MILL TEST REPOET FOR SA105 MATERIAL SHOWS CARBON AT .17 AND Mn AT 1.1. HOW DO WE KNOW IT SHOULD BE FLAGGED OR NOT.”	A mill test report for SA105 showing carbon .17 and manganese 1.1. The poster does not know whether to flag it.	r/Welding, Nov 2023, link
“my boss “took off” an entire high school duct job at 1/8” scale instead of 1/4” scale.”	A drawing scale misread (1/8 vs 1/4).	r/metalworking, Aug 2026, link

Line as written	Reading, and where it comes from	Source
“A decimal point placement on gold plating. 10x too thick”	A decimal point error.	r/metalworking, Aug 2026, link

Three rows (the Ultem ones) are plastic. They are included because the naming problem is the same.

RQ3: Costly mistakes

Claim 3.1. The costliest slips are missed notes and callouts, wrong material or spec, and stale material prices. (15 records in 1 thread, all from the fab-quote thread.) Thirteen commenters in that thread describe separate incidents, so the pattern repeats within the thread, but it is one thread.

Wrong material or grade:

- “Spec and quote was for 316ss sch80 pipe.” (r/metalworking, Aug 2026, score 5) link
- “304ss sch40 was purchased by mistake.” (r/metalworking, Aug 2026, score 5) link
- “it was a $250k job that ended up costing close to $450k.” (r/metalworking, Aug 2026, score 5) link
- “just use the normal 4mm, they will never see it.” (r/metalworking, Aug 2026, score 13) link
- “He misinterpreted the P/O, what the customer actually wanted were shims for those caps.” (r/metalworking, Aug 2026, score 2) link

Missed notes and weld details:

- “Seems like quoting guys don’t understand that the cost between a 3/8” fillet weld and a 1/4” fillet weld can be 3x more.” (r/metalworking, Aug 2026, score 54) link
- “Early in my career I missed a note calling out weld size.” (r/metalworking, Aug 2026, score 54) link
- “Following just the drawings when there were additional written instructions that added way more work.” (r/metalworking, Aug 2026, score 3) link

- “Missed the part where it said the welds are to be purged…” (r/metalworking, Aug 2026, score 10) link
- “They lost money on a big project because of a one liner in the scope.” (r/metal-working, Aug 2026, score 2) link

Units, scale, and numbers:

- “my boss “took off” an entire high school duct job at 1/8” scale instead of 1/4” scale.” (r/metalworking, Aug 2026, score 4) link
- “A decimal point placement on gold plating. 10x too thick” (r/metalworking, Aug 2026, score 17) link

Stale price data in the quoting system:

- “the material pricing in the system wasn't getting updated like I thought it was. Labor was good, material was VERY bad and ended up only making, at best, $2-3k on the whole job. Should have been an easy $15-20k in profit.” (r/metalworking, Aug 2026, score 6) link

Hardware with minimum order quantities and long lead times:

- “Not paying attention to hardware requirements and quoting a couple cents each assuming that is going to cover it then finding out there is a 10,000 piece MOQ (or worse)” (r/metalworking, Aug 2026, score 9) link

Claim 3.2. No gauge-to-thickness error story appears in the 15 threads read. The only gauge discussion (r/sheetmetal) is about whether a 26 gauge 301 part can be bent to a sharp angle. No poster reports a gauge table mistake. The round 1 search listings were not read thread by thread, so this is a finding about the pages read, not about Reddit. The domain-rules track covers gauge tables.

RQ4: Hidden requirements

Claim 4.1. Buyers expect certs and traceability that cost real money, and the paper-work is the product. (16 records in 4 threads: 16 records in 4 threads.)

- “Those certs are what you’re paying for.” (r/Machinists, Jul 2024, score 107) link
- “The are the same bolt. One has certs and one doesn’t.” (r/Machinists, Jul 2024, score 1) link

- “It's not military "grade".... it's military "spec".” (r/Machinists, Jul 2024, score 6) link
- “MIL spec requires full traceability.” (r/Machinists, Jul 2024, score 6) link
- “it’s the documentation, certification, record keeping” (r/Machinists, Jul 2024, score 29) link

Claim 4.2. "DFARS" and "made in the USA" get used loosely, and the bottleneck is what the mill test report says. One thread, low scores.

- “Looking for a supplier for DFARS Compliant 60 series aluminum. Also has to be made in the USA” (r/Machinists, Aug 2023, score 1) link
- “my customer still wants domestic source and will not take anything else” (r/Machinists, Aug 2023, score 1) link
- “Aluminum is exempt from DFARS, so technically, there is no such thing as DFARS compliant aluminum” (r/Machinists, Aug 2023, score 2) link
- “Domestic is doable, just having the CMTR state DFARS compliant is the hard part.” (r/Machinists, Aug 2023, score 2) link

Commenters in the military-fastener thread tie "military" to domestic supply, and one supplier says domestic material alone can cost more than an overseas finished part:

- “Military grade means no Chinese vendors anywhere in the process.” (r/Machinists, Jul 2024, score 3) link
- “their cost from overseas (complete) was cheaper than just the material cost for me” (r/Machinists, Jul 2024, score 1) link

Distributors a commenter named for domestic aluminum (Aug 2023):

- “Some vendors in my area are EMJ, Clinton Aluminum, Yarde, and Ryerson.” (r/Machinists, Aug 2023, score 4) link

Claim 4.3. Checking a mill test report against the spec is done by hand, and the answer depends on which edition of the spec you hold. One short thread.

- “MILL TEST REPOET FOR SA105 MATERIAL SHOWS CARBON AT .17 AND Mn AT 1.1. HOW DO WE KNOW IT SHOULD BE FLAGGED OR NOT.” (r/Welding, Nov 2023, score 1) link
- “I'm using ASTM A105 and my version is different than yours (ASME?).” (r/Welding, Nov 2023, score 2) link

- “can we pass this MTR according to the code(ASME) or flag it, though the material might be okay ?” (r/Welding, Nov 2023, score 1) link

Claim 4.4. Export control can be triggered by the part's use, not by the material line. Single comment.

- “If this is a munitions part this material may be ITAR/ export controlled and you may want to remove this info just as a CYA” (r/Machinists, Apr 2026, score 1) link

Claim 4.5. Buyers want exceptions and assumptions (spec revision, substitutions, test and cert assumptions) visible next to every line. Buyer-side, from the account that started the thread.

- “I like forcing an exceptions column next to every quoted line, not buried in footnotes: spec revision, excluded accessories/packaging, substitute material, test/cert assumptions, warranty/remedy limits, and any Incoterm or freight exclusions.” (r/procurement, Aug 2026, score 1) link

What triggers each requirement, as far as the posters say:

Requirement	What triggers it (per posters)	Source
Mill test report or cert with traceability back to the material	Military or aerospace spec ("MIL spec") on the part; the bolt example	r/Machinists, Jul 2024, link
Domestic source	A customer saying so; the poster does not always say which rule	r/Machinists, Aug 2023, link
DFARS statement on the cert	A defense end use; unclear whether aluminum is covered (one comment says no)	r/Machinists, Aug 2023, link
ITAR or export control	The part is a munitions component	r/Machinists, Apr 2026, link
Spec revision and edition	A drawing that says "per your drawing" without a revision	r/procurement, Aug 2026, link; r/Welding, Nov 2023, link
Notes on the print (weld size, purge)	Written notes that sit apart from the geometry	r/metalworking, Aug 2026, link; r/metalworking, Aug 2026, link

What this says about the four candidate angles

Section 4

Counts come from `voc_records.jsonl`, grouped by the `rfq_theme` field. They are counts of quoted records and distinct threads in this corpus, not a survey.

Angle	What Reddit says	Strength	Caveats
The quote shows its work	Buyers want to see where each field came from and what was assumed (3 records in 1 thread). A supplier-side poster says reviewing is easier than writing from scratch. Another quoted a job of almost $100k with a material price the system had not updated, and kept $2-3k instead of an expected $15-20k; a visible price source and date would have exposed it. Readers distrust AI that cannot be checked.	Strong in logic. Thin in count.	The provenance quotes come from an account that discloses working on quote tooling. The stale-price story is one commenter. Trust issue is one thread.
One-click buyer questions	Shops ask "What angle is needed?", buyers leave out dimensions, and cold RFQs get ghosted or answered with a question. The buyer-side thread says ask for the exact missing field. (8 records in 5 threads)	Moderate	Asking is described as normal and manual. No one says the question step is slow; they say it is skipped or ignored.
Learns the buyer's shorthand	One poster says every RFQ feels like starting from scratch (a single comment). Items show naming variation (Ultem, ULEM, PEI; HRPO vs A36; "DFARS" vs "domestic"). No poster describes one buyer writing the same shorthand repeatedly.	Weak	Single anecdote about rework. The naming evidence shows variation across people, which is a different problem from one buyer's habits.
Catches hidden requirements	Certs and traceability cost money (5 records in 1 thread). DFARS and domestic get conflated (7 records in 2 threads). MTR checks are manual and depend on the spec edition (3 records in 1 thread). Missed notes cause the biggest losses (5 records in 1 thread).	Moderate to strong	The cert evidence is from machinists and inspectors, not from distributor quoting reps. The missed-note stories are about notes on the print, not about requirements the spec only implies.

Possible fifth angle the evidence suggests. Combine the strongest parts of two angles: a quote that lists, next to each line, what it assumed and what it read from the notes on the print (weld size, purge, spec revision, test and cert assumptions), with the source shown. The missed-note stories (5 records in 1 thread) and the excep-

tions-column quote point to it. The same evidence also names a second candidate: a price source and age shown on every material price, from the stale-price story. Both are single-thread evidence.

Ranking from Reddit alone, with confidence. This is my judgment from the counts above.

- 1. Catches hidden requirements, broadened to include notes on the print.
- 2. The quote shows its work.
- 3. One-click buyer questions.
- 4. Learns the buyer's shorthand.

Confidence is low to moderate, because the people closest to RudCo's reps are missing from this corpus and most of the strong evidence is one thread. The open-web track should confirm or overturn this using distributor reps.

Failure mode notes

Section 5

- **Tool-builder posts.** 13 flagged items in `vendor_flags.csv`. The most promoted threads are the repeat-RFQ, quote-format, and quote-comparison threads. The poster of the quote-comparison thread links their own tool. One commenter in the quote-format thread says outright that the poster is fishing for an app.
- **Self-reported anecdotes.** Dollar amounts and outcomes in the fab-quote thread cannot be verified.
- **Survivorship.** The posters are people who stayed and answer questions online. Reps who quit or lost accounts are absent.
- **Recency.** The costly-error thread is from Aug 2026. The cert thread is from Jul 2024. The DFARS thread is from Aug 2023, so it may not reflect current rules.
- **Individual makers.** Three of the 15 threads (HRPO vs A36 for a welding table, the one-off turned part, the 301 stainless part) come from individual makers or founders, not industrial buyers.

Corpus coverage

Section 6

- Subreddits read: r/procurement, r/supplychain, r/sales, r/Machinists, r/metalworking, r/Welding, r/sheetmetal, r/CNC. Round 1 also pulled search pages from r/manufacturing, r/MetalFabrication, r/engineering, r/MechanicalEngineering, r/estimators, r/purchasing.
- Date range: 2023-08 to 2026-08 (quoted items).
- Threads: 15 read (15 of 15 round-2 threads, none empty); round 1: 24 search pages with results, 1 empty (r/purchasing). Per-thread table in `coverage_manifest.md`.
- Records emitted: 94 in `voc_records.jsonl` from 15 threads, dry-run validated, rejected 0.
- Vocabulary note: the shared warehouse vocabulary has no RFQ themes. Each record carries a generic placeholder theme so it validates, plus a real `rfq_theme` field and an `rfq_questions` field. Counts in this file come from `rfq_theme`. If these records are loaded into the warehouse, the RFQ themes need adding to `~/.claude/voc/schema.sql` and `load.py` first. The side field maps "trainer" to the supplier or quoting side and "client" to the buyer side.
- Suggested next pulls (not run; deadline): the "Anything better than Xometry?" thread in r/CNC, and the job-shop quoting threads in r/CNC and r/Machinists, if the open-web track finds Reddit useful.

Appendix 3: Voice of User: Other Sources

Read this first

Track: voice of user from sources the open-web and Reddit tracks did not try. Run 2026-10-10. Covers software review sites, job postings, podcasts, Hacker News and founder posts, employee reviews, and surveys. No Reddit pages were fetched.

The answer. The best new evidence is job postings and one distributor case study. Software review sites gave little, and no podcast, Hacker News thread, or survey gave usable first-person quoting voice. What the new sources add:

Job postings show the steps and the checks. Two postings (Precision Steel Services and Nox Metals) spell out what a quoting rep does: read the request, confirm dimensions and grade, catch special requirements (testing, inspection, tolerances, origin restrictions, cert requirements), document assumptions, and review every quote before it goes out. They are the closest thing found to a metals quoting rep describing the job. They are employer wording, not rep wording.

One named inside sales manager speaks in his own words. Travis Bryant of Puget Sound Pipe & Supply (a pipe, valve and fitting distributor, not a metals service center) says in a vendor case study that a 100-line bid took an hour to an hour and a half, and that he has typed some items "tens of thousands of times" and still made mistakes. It is the only first-person rep voice in this track, and the vendor (Canals) published it.

Customers' shorthand and approved lists are real. The same case study says requests arrive as Excel files full of shorthand and as photos of handwritten lists, and that oil and gas customers required approved product lists and preferred domestic items.

Review sites say little about quoting. The one clear signal is Epicor Prophet 21 (a distribution ERP): TrustRadius's summary of 29 reviews lists an outdated, inefficient quoting process as a con.

No hit-rate or quote-turnaround survey exists in what I could read. The NAW/MDM report is paywalled. The one buyer survey I found (Paperless Parts, 2022) is from a software vendor and I saw it only in a search summary.

Gaps, stated up front.

G2 and Capterra block automated browsers. Capterra showed a bot-check page and G2 returned no text. I did not try to get past either. I read G2 reviews through SaaSworthy, which republishes them and labels the source as G2.

No reviews exist on the pages I could reach for Enmark/ENITEO, Invera, Infor, Eclipse, Proton, Mercura, Hexa or Avent. Details in the coverage table.

No podcast or video transcript was readable. Show notes only.

No Hacker News thread had a founder relaying what metals or industrial distributors told them.

Employee reviews on Indeed mention inside sales but nothing about quoting.

How to read the evidence labels.

Read directly means I read the page text in a browser. Words in quotation marks from these pages are exact.

Via fetch tool means a summarizing tool returned the page content. I paraphrase these and do not use quotation marks, except where noted.

Search summary only means I saw the item only in a search result snippet. It is a lead, not a verified fact.

Vendor means the publisher sells quoting or ERP software. **Secondhand** means a founder, vendor or writer is relaying what customers said.

Contents

Coverage: what I tried, what I expected, what I got

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Question 5: Existing tools, from user voice

Question 6: Price benchmarks

What this evidence says about the four angles

Sources I read but did not use

Cleanup needed

Coverage: what I tried, what I expected, what I got

Section 1

Source type	Where and search terms	What I expected	What I got
Review sites: metals ERP	"Enmark metals ERP reviews quoting G2 Capterra"; SoftwareConnect page for ENITEO	Reviews of ENITEO (Enmark and Bayern Software merged) mentioning quotes	SoftwareConnect page says no reviews have been submitted (read via fetch tool). Search found no G2 or Capterra listing for Enmark. The G2 and Capterra results were for Aptean Metals ERP, one review, not read.
Review sites: distribution ERP	TrustRadius, GetApp UK, SaaSworthy (G2 mirror) for Epicor Prophet 21	Complaints about quoting from reps	TrustRadius summary names a quoting process con (read directly). Two G2 reviewers say order entry is slow (read directly). No metals reviewer.
Review sites: quoting tools	SaaSworthy (G2 mirror) and GetApp for Paperless Parts	Praise and complaints on quoting	Nine G2 reviews read directly, all from job shops and manufacturers, not service centers. Two mention raw material or outside services costs.
Review sites: AI quote tools	Searches for Proton, Mercura, Hexa, Avent reviews	G2 or Capterra reviews	None found. Only vendor pages and Y Combinator launch posts.
Review sites: Infor, Eclipse, Invera	Search "Epicor Prophet 21 reviews quoting"; Eclipse appears only as the ERP at a Canals customer	Reviews with quoting detail	No usable reviews found. Not read.
Job postings	Searches: "metals service center inside sales quoting RFQ", "mill test reports MTR inside sales steel", "quoting specialist estimator metals distributor RFQ certifications"; fetched Boone Steel, A.M. Castle, Lapham-Hickey, Precision Steel Services, Nox Metals, Vulcan, Valley Iron	Step lists and tool names	Strong. Precision Steel Services and Nox Metals read directly and are detailed. Others are thinner.

Source type	Where and search terms	What I expected	What I got
Podcasts and video	MDM podcast, Distribution Talk, MSCI, Metal Center News, Online Metals "Metalpress" podcast	Transcripts or detailed notes on quoting	None readable. Online Metals page returned 403. Distribution Talk and MDM episode notes found were about AI, CRM, and agents in general, with no RFQ detail. Show notes only.
Hacker News and founder posts	Hacker News search (Algolia) for RFQ, distributor, steel, quoting, mill test reports; Y Combinator launch pages for Hexa and Avent	A founder relaying customer pain	Hexa and Avent launch posts (vendor, secondhand). Three HN comments, one relevant buyer comment.
Employee reviews	Indeed reviews for Ryerson and Reliance; Glassdoor search results	Complaints about quote workload	Indeed Ryerson page read directly. Two inside sales reviews, none on quoting. Glassdoor pages returned 403. Reliance page had no inside sales reviews in the filter.
Surveys	NAW, MDM, MSCI, "survey quote response time hit rate"	Quote turnaround or hit rate with sample size	NAW/MDM MarketPulse (about 300 respondents, Baird) is paywalled and not about quotes as far as the listing says. Paperless Parts 2022 buyer survey seen in search summary only. A vendor case study quotes one exec on response time.

Question 0: The job steps, who does them, where time goes, what a better quote wins

Section 2

The steps, as two job postings describe them (read directly)

Precision Steel Services (Toledo, OH; a steel supplier since 1974, per its own posting; posting does not show a date): Estimator - Steel Plate Quoting & Pricing. The "What You'll Do" list is, in order:

1. "Review customer requests, emails, prints, and specifications to fully understand what the customer is asking for"

2. "Verify that quote information is complete and accurate before pricing: dimensions, grades, quantities, and details"
3. "Identify special requirements such as testing, inspection, tolerances, or origin restrictions, and confirm them with the shop before quoting"
4. "Review customer history and win/loss patterns to set a competitive price position"
5. "Evaluate how parts fit standard stock sizes and account for unusable remnant and waste"
6. "Build and verify pricing in our pricing software, including material cost, markup, and processing costs such as burning"
7. "Recognize material we don't stock or processes we can't run in-house, and route those quotes correctly"
8. "Spot jobs that exceed volume, weight, or dollar thresholds and follow the defined escalation path"
9. "Document assumptions, lead times, and any changes clearly on the quote so the customer is never surprised"
10. "Run a final review of every quote for accuracy, minimums, and total-cost sanity before sending"

The same posting then says who hands off to whom: the estimator passes "value-add and time-critical information to Account Management so they can win the order." It also defines success: "Special requirements are verified and written on the quote before it is sent."

Nox Metals (Detroit; aluminum plate, bar and rounds for aerospace and defense; the posting is closed): Sales Development Representative (SDR). Nox calls itself a technology company, so this is a software-forward shop, but the duties are the same job. The listed duties:

- "Own inbound RFQs and quoting end to end, every quote out the door fast and accurate"
- "Translate customer drawings, specs, and POs into clean, complete jobs in NOX NEST"
- "Push orders into production with every detail right, alloy, temper, thickness, tolerances, cert requirements, ship date, packaging"

- "Catch issues before they become problems, missing specs, mismatched POs, unrealistic dates, conflicting cert requirements"
- "Always ask questions, never guess when something is unclear"
- "Capture every quote, every win, every loss in our CRM and quoting system, the data is the moat"

Who does each step

- At Precision Steel, the estimator does intake through the final check. Account Management makes the follow-up call, and "the shop" confirms special requirements and capability. Source: the posting above.
- At Nox, one inside rep owns the RFQ from intake through production handoff, then keeps the customer relationship "inside the four walls." Source: Nox posting above.
- Other postings (via fetch tool, thinner): A.M. Castle's inside sales role includes quoting prices and committing to delivery dates and checking material availability and conformance to customer requirements (jobvite). Boone Steel's inside sales role takes requests by phone, email and counter and develops pricing (betterteam). Vulcan, an Australian metals distributor, has an estimator and internal sales role that quotes from CAD/PDF drawings and supplies test certs (Workable). Valley Iron's inside sales role asks reps to review all quotes and orders for accuracy before sending (InterviewPal).
- A search summary reports that an Aero Metals Alliance (Sunshine Metals) inside sales posting in Wichita sets a goal of answering each RFQ within 24 hours. I could not open the posting (KansasWorks returned an error), so treat this as a lead: KansasWorks search result.

Where the time goes

- **Large bids at a pipe distributor.** A quote in the Canals case study for Puget Sound Pipe & Supply (an inside sales manager, per the case study) says a 100-line-item bid, "even with inventory items and shortcuts," took "an hour to an hour and a half." The page shows this as an unnamed blockquote inside a story about Travis Bryant, Inside Sales Manager, so I attribute it to the case study. The same page says the team aimed for 15 to 30 minute turnarounds before the tool, unless a request was complex. Vendor case study, read directly: Canals.

- **Ryerson, a large US steel service center.** Per Concurrency, the firm that built Ryerson's AI quote tool, Ryerson's "600 sales professionals" handle "more than 100,000 SKUs" and "receive an email request for a quote every 4.5 minutes." Quotes were prepared "by hand, drawing from several systems," and "Time-to-Quote was typically hours and often required multiple clarifying communications between the customer and Ryerson's sales team." Vendor writing about a customer, no Ryerson person quoted, 2023: Concurrency (read directly).
- **An industrial seals distributor.** Concurrency's other case study says manual quoting took over 3 hours per quote, with about 20 quote requests a day arriving through several channels. Vendor writing, no customer quoted, read via fetch tool: Concurrency.

What a faster or better quote wins, and who says it

Value prop	Who says it	Source and label
Speed wins the order	Chief commercial officer of an unnamed national industrial automation distributor, relayed by the vendor Instalily	Paraphrase: he says the chance of winning is higher if you reply within an hour and drops after two, and that past four or five hours it is not worth responding. His firm's median reply time before the tool was three to four hours. Vendor case study, secondhand, via fetch tool: Instalily.
Speed wins the order	Unnamed sales leader at a heavy-duty truck parts distributor, relayed by Proton (vendor)	Paraphrase: the first complete, priced bid usually wins, even at a slightly higher price. Via fetch tool: Proton blog. The open-web track already covers Proton's other claims.
Speed wins the order	Concurrency writing about Ryerson	"Buyers are often impatient, moving on to another supplier if a quote isn't immediately available." Vendor writing, read directly: Concurrency.
Follow-up wins orders	CloudForge CEO Ben Marans (vendor)	"Currently service centers, on average, follow up on 10 to 15 percent of quotes—and that's done manually." Vendor claim, no source given for the figure. Read directly: Modern Metals, July 2024.
More time for follow-up and selling	Puget Sound Pipe & Supply staff	In the case study, follow-up "always came last" before the tool (unnamed blockquote), and the team cleared a product line that had waited six years. Vendor case study, read directly: Canals.
Fewer errors	Travis Bryant, Inside Sales Manager, Puget Sound Pipe & Supply	"I've typed up some of these items tens of thousands of times—and still made mistakes." Vendor case study, read directly.
Less dependence on senior staff	Same case study	Newer reps "had to lean on senior folks just to figure out substitutes or where to look" (unnamed blockquote; vendor case study, read directly).
Quote win rate up	Chris Krebs, president of Fighter Jet Metals, a Florida service center, quoted by CloudForge	"CloudForge has been a gamechanger for us—dramatically increasing our quote win rate by over 50 percent and cutting quote-response times in half." Vendor-published customer quote, read directly: Modern Metals.
Fewer sales lost to an unresponsive team	Hexa founders (Y Combinator launch post)	Paraphrase: they say slow quoting costs distributors more than 15 percent of revenue a year. Vendor claim, via fetch tool: Y Combinator.

Which value props reps and employers name themselves. In the two detailed postings, "accurate the first time" and "fast" are listed together. Precision Steel's success list says quotes should go out "faster than the customer expected, with no compromise in quality" and "accurate the first time, without preventable corrections." Nox lists "every quote out the door fast and accurate." Margin appears only at Precision Steel (competitive and profitable pricing, waste and remnant "covered in the price, not absorbed by the company"). Neither posting mentions win rate as a goal for the estimator; both treat winning as the account manager's job after a good quote.

What I did not find. No service center publishes a quote hit rate or turnaround benchmark here. The Aero Metals Alliance 24-hour goal is the only stated RFQ turnaround in a posting, and I could not open that page.

Question 1: Pain in their own words, and buyer annoyance

Section 3

Reps and employers (first-person where marked)

- **The only first-person rep voice found.** Travis Bryant, Inside Sales Manager at Puget Sound Pipe & Supply (Burlington, WA; over a decade with the company), in a Canals case study: "I've typed up some of these items tens of thousands of times—and still made mistakes." On the new tool: "You gotta train it. You gotta show it what it needs to do before it'll do it for you." And on backlog: "There was a time when we had a backlog of quotes that was a week out." Vendor-published, read directly: Canals. He is a pipe, valve and fitting distributor, not a metals service center. Canals is the vendor and chose what to publish.
- **Employer wording about the pain, from postings.** Precision Steel: "This is not just math. It requires print-reading skill, sound judgment, attention to detail, and the ability to work quickly through a high volume of requests without letting errors reach the customer." Read directly: posting. The posting also asks for a "methodical, repeatable process instead of relying on memory" and says "You never guess on pricing or capability. You verify it."
- **Nox on the day-to-day.** The posting asks for someone "organized enough to juggle dozens of open quotes and orders without dropping one" and "precise under pressure and reliable, customers count on the dates and prices you commit to." Read directly: Nox posting.

- **Service center ERP is dated, per its sellers.** CloudForge CEO Ben Marans: "A lot of software running at today's service centers was designed in the 1970s and '80s. It hasn't evolved and is rife with tech-debt." Paragon Tube CEO Phil Henry, quoted by CloudForge: "We have stuck with the same antiquated software for almost two decades." Both read directly: Modern Metals. The first is a seller's claim. The second is a customer quote chosen by that seller.
- **Plymouth Tube (a tube maker, Aptean case study).** A search summary says customer service reps built quotes by hand from price sheets and undocumented know-how, and that the process frustrated both the company and customers, especially when a backup rep covered for someone out. This is vendor text. I could not open the page: search result.
- **Founders (secondhand).** Avent founder Abhay Kalra, in a launch post, says growth in distribution is capped by hiring, that critical knowledge lives in inboxes or people's heads, and that training new reps takes months. He describes his own observations as an operator. The post does not relay customer quotes. Via fetch tool: Y Combinator launch.

Employee reviews (Indeed, Ryerson, read directly)

Nothing about quoting. The inside sales reviews on Indeed are two:

- Inside Sales Representative, March 11, 2026: "There is a lot of information to learn in the metals space. I also found inside sales to be a little boring/monotonous. There was not a lot of cold calling."
- Inside Sales Representative, Carrollton, TX, November 9, 2025: "Constant operational and shipping problems that lead to losing customers." The same review says the team was "Super understaffed."

The first supports the "steep learning curve" point (a lot of product knowledge to learn). The second is about operations, not quotes. A third, unlabeled "Mid" reviewer on the main page (June 25, 2026) says "You get a book of business so normally you aren't making cold calls."

Buyers

- **Almost nothing first-person.** One Hacker News commenter (auxym, January 2023, a mechanical engineer buying prototype parts in Canada) says the only way to

learn industrial supplier prices is to call or email an RFQ and wait a day or two. Via fetch tool: HN comment. It is a buyer describing waiting, from one person.

- **A buyer survey (search summary only).** A search summary of the Paperless Parts 2022 Custom Part Buyer Survey (more than 600 US manufacturing professionals; vendor survey, on custom machined parts) says 47 percent said a shop's response speed directly affects whether they work with it, 69 percent look to capabilities, and 54 percent look to quality certifications as a sign a shop will deliver. I could not open the article: Industrial Distribution.

Question 2: Messy lines (what these sources show)

Section 4

These sources describe the mess but give no verbatim line items.

- **Canals case study (read directly).** Requests come as "Excel file[s] filled with shorthands" and "a picture of a handwritten list." Bids can run 100 or more lines. The page also says the catalog was messy ("despite a messy ERP catalog"). Source: Canals.
- **Nox posting (read directly).** Problems to catch: "missing specs, mismatched POs, unrealistic dates, conflicting cert requirements." It also says reps translate "customer drawings, specs, and POs" into jobs.
- **Ryerson (read directly).** Requests "came largely by email with or without attachments, and often by phone." Source: Concurrency.
- **eAlliance steel bar case and the USAF and USACE RFQ lines** are in the open-web track. Use those for the test set.

Question 3: Costly mistakes

Section 5

This track found no new documented cost of a reading error. What the new sources show:

- **What employers tell reps to check, which implies where mistakes happen.** Precision Steel lists dimensions, grades, quantities, special requirements, remnant and waste, material the shop does not stock, and thresholds that need escalation. Waste and remnant are to be "covered in the price, not absorbed by the

company," which says an unpriced remnant costs the supplier margin. Source: posting (read directly).

- **Wrong picks at junior level.** Puget says newer reps relied on seniors "to figure out substitutes" (unnamed blockquote, vendor case study).
- **Quote estimates that are a little off.** A G2 reviewer of Paperless Parts, Christopher K (May 16, 2023): the cycle time estimate "is a little off at times for larger more complex parts" and outside services such as black oxide "arent properly calculated based on the weight of the parts." Read directly through SaaSworthy: SaaSworthy. This is a job shop, and it is about pricing, not reading the request.

Question 4: Hidden requirements

Section 6

The job postings treat hidden requirements as part of the rep's job. None says how often they are missed.

Requirement	Trigger or note	Source
Testing and inspection	Rep must "identify special requirements such as testing, inspection, tolerances, or origin restrictions" and confirm with the shop before quoting	Precision Steel posting, read directly
Origin restrictions	Same line	Same posting
Cert requirements on the order	Nox lists cert requirements next to alloy, temper and thickness when pushing an order to production; flags "conflicting cert requirements" as an issue to catch	Nox posting, read directly
Aerospace and defense paperwork	Nox lists "AS9100, DFARS, AMS, ASTM" as nice to have for the rep	Nox posting, read directly
Mill certs and MTRs	Lapham-Hickey requires the ability to read and interpret mill certifications and material test reports	JobTarget posting, via fetch tool
Test certs supplied with the quote or order	Vulcan's role includes "supplying test certs"	Workable, via fetch tool
Approved product lists and domestic preference	Puget's oil and gas customers required approved product lists and preferred domestic items	Canals, read directly

The domain-rules track covers what triggers each cert. This track adds that employers expect reps to catch them from the request text, which supports the "hidden requirements" angle as a real part of the job.

Question 5: Existing tools, from user voice

Section 7

Epicor Prophet 21 (distribution ERP)

- **TrustRadius, read directly.** TrustRadius's own summary of 29 verified reviews published in the last 18 months lists "Outdated and inefficient quoting process" among cons. The synthesis page says 10 percent of reviewers raised an outdated quoting

process, "described as slow and difficult for sales teams to utilize effectively." This is TrustRadius's machine-written summary of reviews, not a single reviewer's words. Sources: reviews page and review insights.

- **G2 via SaaSworthy, read directly.**
  - Verified user, May 20, 2026: "I don't like the UI, and the overall speed of the system. Mostly in order entry, our reps tend to get slowed down."
  - Verified user, May 19, 2026: "Order entry can be slow at times, causing delays when working with customers."
  - Verified user in Industrial Automation, May 8, 2025, on the upside: "Efficiently service customer orders, from quoting through purchasing, to shipping, to accounting."
  - Source: SaaSworthy. Reviewer roles are mostly not shown. None are metals.
- **GetApp UK, read directly.** No quoting complaints on page one. One reviewer in wholesale: "I do not see a return on our investment." GetApp.

Paperless Parts (quoting for job shops and "metal service providers")

All read directly through SaaSworthy, which labels these as G2 reviews: SaaSworthy. These are job shops and manufacturers quoting made-to-order parts from CAD, not service centers quoting stock.

- **Praise.**
  - Christopher K (May 16, 2023): "great at identifying features of a part that we might overlook while quoting." On volume: "We had a backlog of jobs to quote, which would take forever to do manually."
  - Tony J (Nov 18, 2022): "We can quote parts 10X faster than using our old ERP system."
  - Brian K (Nov 18, 2022): "A key issue we had was finding the time to quote projects, manage the quotes and calculate win rates."
- **Complaints.**
  - Tony J: "Not being able to quote raw material efficiently." This is the closest hit to the RudCo case (raw material), and it comes from a tool that does not target service centers.
  - Christopher K: cycle time estimate and outside services off at times (above).
  - Michael N (July 11, 2022): the downside is monthly subscription fees.

- **Read the praise for trust.** "Identifying features ... we might overlook" is an unprompted reviewer praising a tool for surfacing what the human missed. "A little off at times" is the opposite: a number the user cannot check erodes trust.

Enmark/ENITEO, Invera, Aptean Metals ERP, Infor, Eclipse, AI tools

- **Enmark/ENITEO:** the SoftwareConnect page says no reviews have been submitted. Its listed pros and cons are the site's own, not user words (via fetch tool). Cons listed: "Must request pricing details from the developer" and "Complex set-up and integration."
- **Aptean Metals ERP:** a search shows one G2 review. I could not read it (G2 page returned 403).
- **Proton, Mercura, Hexa, Avent, Canals:** no G2, Capterra or TrustRadius reviews found. Evidence is vendor pages and case studies only. Canals claims it learns each rep's and each customer's way of quoting. The Puget Sound case study quotes the user, not a review site.
- **Ryerson's tool, "Metal Detector":** a custom build by Concurrency on Azure OpenAI, not a product anyone can buy. Concurrency says it handles over 50 percent of quote requests without human intervention and credits it with as much as a 3 percent increase in top-line revenue (as reported by the fetch tool; the page also lists a $150 million estimate). Both are vendor claims about a customer: Concurrency.

Question 6: Price benchmarks

Section 8

Not covered by this track. One sentence from CloudForge's CEO is relevant: the platform "brings actual and real-time replacement costs into the process" so that service centers no longer "rely upon tribal knowledge to guesstimate the cost" (Modern Metals, vendor, read directly).

What this evidence says about the four angles

Section 9

1. The quote shows its work (every field links back to the RFQ words). Strongest support in this track.

- Both detailed postings make the check and the paper trail the center of the job: "Document assumptions, lead times, and any changes clearly on the quote so the customer is never surprised," "Run a final review of every quote," "every assumption documented" (Nox), and "You never guess on pricing or capability. You verify it."
- A Paperless Parts reviewer praises the tool for catching what the human might overlook, and another says its estimated cycle time is "a little off at times." Trust depends on whether the user can see why a number is what it is.
- The Canals case study says the rep's remaining job is "reviewing and confirming item matches." Showing the source words next to each match fits that step.
- Gap: no one in these sources says "I wish the tool showed me where it got this." The support is inferred from the review step employers require.

2. One-click buyer questions. Moderate support.

- Ryerson's quotes "often required multiple clarifying communications between the customer and Ryerson's sales team" (vendor writing). Nox's "Always ask questions, never guess when something is unclear" and Precision Steel's "confirm them with the shop before quoting" both describe asking as a normal step.
- No source says buyers like or dislike being asked. The open-web track found a fabricator (Staub) saying stopping to ask slows quoting. Nothing here adds buyer-side evidence.
- Existing tools: nothing found that does this by name.

3. Learns the buyer's shorthand. Real need, but a tool already claims it.

- Puget Sound's case study describes Excel files of shorthand and handwritten photos, and says the AI "learns how each rep quotes and what specific customers need" (vendor claim). The user quote about "teaching" it comes from a named manager. Avent's founder says knowledge lives "in inboxes or people's heads."
- Risk for uniqueness: Canals (a funded vendor with a published customer story) already sells this for distributors. A judge who knows Canals will not see it as new. A demo that shows the learned reading carrying over to a second RFQ would still be a clear trust moment.
- Also note: the case study says early users "didn't get item matches right away" before it improved. Nothing in these sources measures how much it learns.

4. Catches hidden requirements. Employers name it, no tool review mentions it.

- Precision Steel puts "testing, inspection, tolerances, or origin restrictions" as a rep duty. Nox lists cert requirements, DFARS, AS9100, AMS and ASTM. Lapham-Hickey requires reading MTRs. Puget's oil and gas customers required approved product lists and domestic preference.
- No review, case study or founder post I read says an existing tool flags requirements the buyer did not state. Of the four angles, this is the one with the least vendor coverage in this track. That is weak evidence of a gap, because I only read a handful of vendor pages.
- Gap: no source gives how often or at what cost a cert is missed.

A possible fifth angle, from the vendors' own claims only. CloudForge says service centers follow up on only 10 to 15 percent of quotes (vendor claim, no source) and Puget says follow-up "always came last." Follow-up after the quote is sent is outside the brief's RFQ-to-sent-quote scope, and the only evidence is vendor-published, so I record it as background.

What I would not rely on from this track. All speed-wins-orders numbers come from vendors or from unnamed executives relayed by vendors. The only first-person rep voice is a vendor-published case study about a pipe distributor. The review-site evidence is thin and mostly from job shops or generic distribution ERP users.

Sources I read but did not use

Section 10

- EDN, May 6, 2002 (Newark RFQ software, via fetch tool): electronics distribution, 24 years old, not metals.
- Rotabull "When will I get a quote" (redirect not followed): aerospace parts brokers.
- Gerdau on Microsoft's site: the page quotes three Gerdau staff, none about quoting.
- Kansas Works, Job Bank Canada, and Online Metals podcast pages returned errors; details above come from search summaries and are marked.

Cleanup needed

Section 11

- None. I deleted nothing and created only this file.

- One empty Chrome tab (tab 530229090, "New Tab") was created by the browser tool when it started a tab group. I did not use it and left it open in case another agent shares the group. I closed my own tab.

Appendix 4: Domain Rules: Specs, Gauges and Hidden Requirements

Summary: what this track found

Track Domain rules (research questions 2, 3 and 4 in the research brief). Written 2026-10-10.

The gauge trap is real, checkable, and already in our own product brief. The product brief shows "304 SS sheet 16ga" and "Stainless 304, 0.060 in" as the same item. They are not. 16 gauge stainless is 0.0625 in. 0.060 in matches 16 gauge *mild steel* (0.0598 in). The gap is 4.2% of thickness. A demo that gets this right, and says why, is easy to verify live.

Hidden requirements are real, but most of them are not in the line. Three of the five big ones (DFARS, Buy America, ITAR) are triggered by the *contract, the project, the ship-to, or an attached drawing*. The grade only decides whether the rule bites. The one real public metals RFQ I found (State of Iowa, line numbers 1 to 49) names no certification or metal origin requirement on any line. Its only origin wording is a general Iowa-products preference for tied bids. A tool that flags hidden requirements must read the whole RFQ package, and must stay quiet when nothing triggers.

Some requirements can be read straight off the line. A spec prefix or word often implies paperwork: `SA-240` (ASME code material), `A500` (structural tube that ships with a mill test report), `304/304L` (dual chemistry), `NACE`, `DFARS`, `Buy America`, `domestic`, `3.1`, `PMI`. These are the safest flags to show first.

The costliest mistakes cannot be fixed with paperwork afterward. Wrong origin (melted abroad), wrong grade, and wrong spec family (A513 vs A500, A53 vs A500) mean the physical metal is wrong. A wrong gauge table is cheaper but is silent: it is a 4.5% to 24% thickness error that still looks like a clean quote.

Public evidence has a ceiling. ASTM and ANSI pages are paywalled or blocked (403). Most spec facts below come from distributors, trade associations, and forums. Each claim carries a confidence tag. I found no public dollar figure for what a misread RFQ line costs a supplier. The ranking in "Costliest reading errors" is my judgment from the consequences, and says so.

Contents

1. How the same item gets written different ways
2. Common specs by product form
3. Hidden requirements: what triggers each one
4. Costliest reading errors
5. What this means for the two decisions
6. Test set
7. Gaps and source quality

Cleanup needed

How the same item gets written different ways

Section 1

1.1 Gauge: the number means a different thickness for each metal

A gauge number is not a thickness. Each material family has its own table. Metal Supermarkets, a metals distributor, publishes one table per material and says: "when measuring sheet product, gauges are neither standard nor metric, and have independent values." Its example: an 8 gauge aluminum sheet is 0.1285 in, an 8 gauge stainless sheet is 0.17187 in (Metal Supermarkets, sheet vs plate).

Gauge values for the common gauges, inches (source: Metal Supermarkets sheet metal gauge chart; tube wall column from Wikipedia, Sheet metal, gauge table):

Gauge	Mild (carbon) steel	Galvanized steel	Stainless steel	Aluminum	Steel tube wall
10	0.1345	0.1382	0.14062	0.1019	0.134
11	0.1196	0.1233	0.125	0.09074	0.120
12	0.1046	0.1084	0.10937	0.08081	0.109
14	0.0747	0.0785	0.07812	0.06408	0.083
16	0.0598	0.0635	0.0625	0.05082	0.065
18	0.0478	0.0516	0.050	0.04030	0.049
20	0.0359	0.0396	0.0375	0.03196	0.035

- **Two published sources agree to the digit** on the mild steel, galvanized, and stainless columns. They differ on one aluminum cell: 14 gauge aluminum is 0.06408 in at Metal Supermarkets and 0.063 in at Wikipedia (a 1.7% gap). Evidence: both links above. Confidence: high for steel and stainless, medium for aluminum.
- **Metal Supermarkets prints a caveat under its charts:** "Actual thickness will vary due to tolerances." Wikipedia adds that the Manufacturers' Standard Gauge "has no inherent tolerances," and quotes ASTM calling the gauge number an "archaic term of limited usefulness not having general agreement on meaning" (Wikipedia). So the table gives the nominal value. The buyer's tolerance is a separate question.
- **Tube is its own table.** 14 gauge tube wall is 0.083 in; 14 gauge mild steel sheet is 0.0747 in. That is 11.1% apart. 16 gauge tube wall (0.065) is 8.7% thicker than 16 gauge sheet (0.0598). The rule must know the *form* (tube or sheet), not only the material.
- **Real suppliers do not always follow the table.** Speedy Metals lists "10ga (0.135") 304 Stainless Steel Sheet" (Speedy Metals), which is 4.0% under the stainless table value (0.14062) and close to the mild steel value (0.1345). The University of Alberta store lists "0.063 (16 Ga.)" 304 sheet, 1/16 in, 2B (UAlberta): the same item written three ways in one listing, with 0.063 rounding the table's 0.0625 by 0.8%. Confidence: high that suppliers differ; I cannot say which supplier is "right."

- **Sizes of the errors** (computed by domain_rules_gauge_calc.py from the table above):

Wrong-table mistake	Thickness error
Stainless read with the mild steel table, gauges 10 to 20	+4.5% to +4.6% (0.0027 in at 16 gauge)
Galvanized read with the mild steel table	+2.8% at 10 gauge, +6.2% at 16, +10.3% at 20
Aluminum read with the mild steel table	-11.0% to -24.2%
16 gauge stainless vs the PRD's "0.060 in"	+4.2%

What the product should do: look the gauge up in a table keyed by material *and* form, show the table name and value on screen, and ask when the material is missing ("16 ga sheet 4x8" with no metal has four candidate answers: 0.0598, 0.0625, 0.0635, 0.0508 in). When the buyer wrote both a gauge and a decimal that disagree under the right table (as in the PRD example), ask which one governs.

1.2 Size notation: feet, inches, and positional shorthand

The same sheet is "4x8", "4' x 8'", "48 x 96", or "48" x 96"". 4 ft x 8 ft is 48 in x 96 in, which is 1219 mm x 2438 mm. 4 x 10 ft is 48 x 120 in. (Computed in the calc script; the 48 x 96 and 4x8 pairing appears in the State of Iowa RFQ, which writes "Cold Rolled Sheets 4 x 10" on one line and "48" x 120" x 14 Gauge" under it.)

Unit marks are dropped, so the meaning of a number depends on the product form. The Iowa RFQ shows it line by line. In "Strips 3 x 20 x 1/8"" the detail line says 3 in wide, 20 *feet* long, 1/8 in thick. In "Structural Angle 3 x 3 x 1/4" the same notation means inches for all three numbers, and the length (20 ft) sits only in the detail line. In "RectangleTubes 2 x 1.5 x 24 x .120", 24 is feet and .120 is a wall in decimal inches. Evidence: Iowa RFQ 245, lines 1 to 10. Confidence: high (one real RFQ, direct text). So the parser needs product-form rules: which number is length, which is thickness, which unit is usual.

Fractions and decimals both appear. The same RFQ writes 1/8" and .120 in adjacent lines. Plate is usually fractional, sheet is usually gauge (Metal Supermarkets).

1.3 Grade notation

- **304, 304L, and "304/304L" are three different requests.** 304 allows carbon up to 0.08%; 304L allows 0.03%; the A240 chemistry rows are 304: 0.08 C max, 18.0 to 20.0 Cr, 8.0 to 11.0 Ni and 304L: 0.03 C max, 8.0 to 12.0 Ni (Saky Steel, A240 grades, vendor table, medium confidence).
- **Dual certified 304/304L is one heat that meets both.** Distributors sell it as "meets specifications for both 304 and 304L" (Speedy Metals). Engineers on Eng-Tips describe it as meeting the mechanicals of straight grade and the chemistry of L grade (Eng-Tips thread, forum, medium-low confidence). An ASME interpretation quoted there allows dual-stenciled plate to be used as either grade at or below 1000 F.
- **Substitution only runs one way.** A line that says 304 *or* 304L can usually be filled with dual-cert stock. A line that says "304L" cannot be filled with plain 304 stock whose carbon is above 0.03. A line that says "304/304L" asks for the dual cert itself. This is my reading of the chemistry limits above; confirm the buyer's spec before substituting.
- **A36 and A572 Grade 50 can be the same plate.** A36 has no upper yield limit, so producers sell one plate certified to both. A36 minimum yield is 36 ksi; A572 Grade 50 is 50 ksi. If the buyer buys it as A36, they design to 36 ksi; to design at 50 ksi it must be certified to A572 Grade 50 (AISC Steel Interchange, July 2014 as summarized in search results; the PDF itself returned 403 to me; supported by Tampa Steel). Confidence: medium.
- **Aluminum temper is part of the grade.** 6061-T6 and 6061-T651 differ: T651 is the stress-relieved form for plate (MakeItFrom, medium-low). A buyer who writes "6061-T6 plate" often means T651; ask.

1.4 Finish

- **2B is the standard cold-rolled mill finish; No. 4 is a brushed finish.** "No. 2B is the mill finish, meaning it has not been processed further." No. 4 has "a pattern of fine parallel lines" and varies between suppliers, so Metal Supermarkets suggests asking for a sample (Metal Supermarkets finishing options). Speedy Metals defines 2B as "cold rolled on polished rolls, annealed, pickled, and skin passed" and #4 as "polished with fine abrasive finish" (Speedy Metals), but its own product list says "#2 Finish" for the same sheets, so even one vendor mixes labels. ASTM A480 is named as the stainless sheet and strip finish standard by a Finishing.com answer, but I did not read its text (Finishing.com, forum, low confidence).

- **Finish is often missing.** The Iowa line "Type 304-2B" gives it; many lines do not. Missing finish on stainless sheet is a question, not a default.

1.5 Form: sheet vs plate vs flat bar

- **The common cutoff is 1/4 in (0.250 in, 6.35 mm).** Plate is 1/4 in or thicker; thinner is sheet. Metal Supermarkets applies this to all metals. Wikipedia says the same: "thicker than 6 mm (0.25 in) are considered plate" (Metal Supermarkets, Wikipedia). Confidence: high for the convention.
- **The standards use other lines.** A forum post quotes ASTM A6 (carbon structural steel) as treating hot-rolled material as plate at 0.230 in or thicker for widths 8 to 48 in, and 0.180 in or thicker over 48 in wide; and notes the sheet spec A568 caps hot-rolled sheet at 0.230 in (Eng-Tips, forum quote of a standard I could not open; low-medium confidence). Other sources use 3 mm (0.118 in) as the line. So a 3/16 in (0.1875 in) item is sheet by the 1/4 in rule and plate by the 3 mm rule. A line in that range needs the buyer's word for what it is.
- **Narrow flat stock is a third category.** A flat 0.230 in or thicker and 6 to 8 in wide is classed as bar in A6 per the same forum and an AISC table cited in search results (secondary; I could not open either). Several Iowa lines are 1 in to 6 in flats; they are bar or flat bar, not sheet or plate.

Common specs by product form

Section 2

Form	Common specs	What the spec tells you	Source and confidence
Stainless sheet, strip, plate	**ASTM A240** (ASME twin: SA-240)	Covers chromium, chromium-nickel, and chromium-manganese-nickel plate, sheet and strip "for pressure vessels and for general applications"; does not by it-self fix grade, finish or cert type	Saky Steel (vendor, medium)
Carbon structural plate and shapes	**A36**, **A572 Gr 50** (A6 is the general-re-quirements spec)	A36: 36 ksi min yield. A572-50: 50 ksi. Often dual certified	AISC via search (medium)
Structural tube (HSS)	**A500** Gr B/C	Round, square, rectangular; physical properties veri-fied; each bundle ships with a mill test report; Grade C yield 46 ksi	Atlas Tube, Steel Tube Institute (manufacturer and trade group, medium-high)
Mechanical tube	**A513**	Tighter dimensions; "no physical requirements or structural verification"; no MTR provided in Atlas's comparison	Atlas Tube (medium)
Carbon pipe	**A53** Gr B	Round only; sold by nominal size and schedule; 21 and 42 ft lengths; hydrotested; pressure and mechanical use	Steel Tube Institute (medium-high)
Austenitic stainless pipe	**A312**	Seamless and welded stainless pipe. I did not find a source for the scope in this session; general industry knowledge	Unverified here
Stainless bar	**A276** (general use, machining) and **A479** (boiler and pressure vessel use)	Same chemistry for 304 and 316 in vendor tables; the choice signals the end use and documentation	Saky Steel, MachineMfg (vendors, medium-low)
Aluminum sheet and plate	**ASTM B209**	"Aluminum and aluminum-alloy flat sheet, coiled sheet, and plate"	ASTM store listing (standards body, scope line, high)

Form	Common specs	What the spec tells you	Source and confidence
Aluminum extrusions (bar, shape, tube)	**B221**	Extruded bars, rods, profiles, tubes. Taken from search summaries; the ASTM page was not reachable	Medium-low

Spec mixing is common in real RFQs. Iowa lines write "Square Tube A513-A500 Grade B" (two different specs, one grade) and "Hot Rolled Strips CQ-A36" (a sheet quality term fused to a structural grade). A500 is structural with verified properties; A513 is mechanical with no property requirements (Atlas Tube). These are different products, so the line is a question, not a reading.

ASME SA vs ASTM A. The `S` prefix means ASME reviewed and adopted the ASTM spec for Boiler and Pressure Vessel Code use. "Material supplied to an SA specification meets the requirements of the ASME Boiler and Pressure Vessel Code, Section II." Some grades are identical; others carry extra ASME requirements. Many mills dual-mark so one piece serves both (Wermac, engineering reference site, medium). Chemistry can differ by edition: a forum shows SA-240 Type 304 chromium at 18 to 20% while a later A240 edition listed 17.5 to 19.5% (CR4 thread via search, low).

AMS (aerospace) specs: not researched in this pass because AMS documents are paid SAE standards and no public source came up. Treat an `AMS` number on a line as a trigger for "aerospace, ask for certs and origin" until someone with access checks.

Hidden requirements: what triggers each one

Section 3

3.1 Summary table

Requirement	Triggers inside the line	Triggers outside the line	Confidence
Mill test report (MTR)	Spec families that ship with one (A500); "MTR", "certs", "cert", "3.1"; pressure and code grades (A240 for vessels, A312, A479, SA-xxx)	Buyer's PO terms, project specs, code work; service-center defaults	High
EN 10204 3.1 / 3.2	"3.1", "EN 10204", "3.2", "witnessed"	Export orders, EPC and oil and gas buyers	Medium-high
DFARS 252.225-7009 (specialty metals)	"DFARS", "specialty metals", "melted in USA", "qualifying country"; chemistry that is a specialty metal (any stainless, nickel alloy, titanium)	Defense contract flow-down; defense-contractor ship-to; DPAS rating (DO or DX) on the PO	Medium-high
Buy America / BABA	"Buy America", "domestic", "American made", "23 CFR 635.410", "melted and manufactured in USA"	Federal or state-DOT-funded project; water or transit infrastructure grants	Medium-high
Country of origin / melt-and-pour	"Domestic only", "no China", "melted and rolled in USA", "mill origin"	Government buyer, project spec	Medium
ITAR	"ITAR", "export controlled"; rare in raw metal text	A drawing or spec attached with an ITAR marking; defense program name	Low-medium
PMI (positive material identification)	"PMI", "100% PMI", "XRF verified"	Refinery, chemical, high-temperature piping, sour service	Medium
NACE MR0175 / ISO 15156	"NACE", "MR0175", "sour service", "H2S", "HIC"	Oil and gas, refinery project	Medium

Requirement	Triggers inside the line	Triggers outside the line	Confidence
ASME SA (code material)	`SA-240`, `SA-312`, `SA-516`, "ASME", "code material", "U-stamp"	Pressure vessel or boiler fabricator as the buyer	Medium-high

3.2 Detail, one requirement at a time

Mill test reports (MTR). An MTR reports one heat's chemistry and mechanical tests; whether the buyer gets one is set by the order, not by the metal.

- AISC states the content of an MTR for hot-rolled plates and shapes comes from ASTM A6 Section 14: heat number, grade, sizes, tension and chemistry results. Whether one must be furnished is set by the purchase order or project spec (AISC Steel Solutions Center, AISC Steelwise Aug 2014, both as summarized by search; the AISC pages returned 403 to me).
- Service centers differ on the default. Ryerson's instructions to its own suppliers say: "Mill test reports are required on all products unless otherwise specified in the Purchase Order" (Ryerson M200, medium). So for RudCo the real question is whether it *keeps heat traceability on cut pieces*, not whether a certificate exists somewhere.
- A500 tube ships with an MTR per bundle; A513 does not (Atlas Tube). A buyer who writes `A513` may expect no cert; one who writes `A500` expects one.

EN 10204 certificate types. 3.1 is the usual request for critical material; it means test results from the delivered lot, signed by the manufacturer's independent inspector.

- 2.1: compliance statement only, no test results. 2.2: manufacturer test results from non-specific inspection (not tied to your lot). 3.1: specific test results from the material supplied, signed by an authorized inspector independent of production. 3.2: 3.1 content plus validation by the buyer's representative or a designated third party (Material Welding, Project Materials; vendor and blog sources agree with each other; the standard itself is paid). One of them contradicts itself on who issues a 3.1, so confirm with the buyer's spec. Confidence: medium-high on the four types.
- 3.2 involves a third-party inspector and can add scheduling, so it is a flag that needs a human, not an auto-quote.

DFARS 252.225-7009 (Restriction on Acquisition of Certain Articles Containing Specialty Metals). The clause says specialty metals in delivered items must be melted or produced in the United States, its outlying areas, or a qualifying country. The grade decides whether the rule *can* apply; the contract decides whether it *does*.

- Specialty metals are defined by chemistry: steel with more than 0.25% of aluminum, chromium, cobalt, molybdenum, nickel, niobium, titanium, tungsten, or vanadium (or above set manganese, silicon, or copper limits); nickel and cobalt alloys above 10% other alloying metals; titanium and zirconium alloys (DFARS 252.225-7009 on acquisition.gov). By that definition 304 and 316 stainless (18% chromium, per the chemistry in section 1.3) are specialty metals; ordinary A36 plate is generally not (my reading; chemistry check needed).
- The clause must flow down to subcontracts, including commercial ones, for items containing specialty metals (paragraph (e)). So a distributor gets the rule from its customer, not from the government.
- Exceptions exist (commercial off-the-shelf items, a 2% minimal-content rule, a government availability determination), each with conditions. "Specialty metal mill products" are carved out of the COTS exception: bare sheet, plate and bar do not get that exemption.
- Triggers outside the line: a defense-contractor ship-to, a DPAS rating (a "DO" or "DX" symbol) on the PO, a government contract number. A DPAS-rated order carries its own duties: written acceptance or rejection within 15 working days for DO and 10 days for DX, and the rating must pass to the supplier's suppliers (15 CFR 700.13, DLA DPAS brochure, summarized by search).
- Confidence: high on the clause text (primary source via fetch). Medium on the COTS detail.

Buy America / Build America, Buy America (BABA). For federally funded highway work, steel and iron must have all manufacturing processes, including coating, in the United States, and the mill test report must say where it was melted and manufactured.

- The federal rule: no permanently incorporated iron or steel, or all manufacturing processes, "including application of a coating," must occur in the US. Coating covers galvanizing, epoxy and painting; the coating *material* is not covered (23 CFR 635.410 on eCFR, FHWA Buy America Q&A, both via search summaries). If any bending, drilling, or coating happens abroad, the product becomes foreign.

- State DOTs add paperwork rules. Caltrans says mill test reports must state where steel was melted and manufactured. Montana DOT rejects a bare "meets Buy America" and wants wording like "Melted and Manufactured in the United States" or "Complies with 23 CFR 635.410" (Caltrans, Montana DOT form).
- BABA (federal infrastructure grants beyond highways): EPA's FAQ says the key step is the initial melting and pouring, which must be in the US, and raw materials need not be domestic (Maine SRF reproduction of EPA FAQ). Thresholds in these state and agency documents (0.1% or $2,500 minimal use; 50% or 55% manufactured products) differ by document, so I do not list them as rules.
- Confidence: medium-high on the core rule; these are agency summaries, not me reading the regulation in full.

Country of origin and "melted and manufactured." Two different questions get blended in RFQs. **Where was it melted and poured** (the furnace), and **where was it rolled or finished**. DFARS asks about melting or producing; Buy America asks about all manufacturing steps including coating. A buyer who writes only "domestic" has not said which. This is a clarifying question, not a guess.

ITAR. Raw metal is rarely ITAR itself; the controlled item is usually the drawing or technical data attached to the RFQ. I found no public source that settles whether plain mill-form titanium or stainless is on the US Munitions List. The sources I found describe ITAR as covering articles specially designed for military or space use and their technical data (voestalpine ITAR program, Performance Titanium export FAQ, which points to the Export Administration Regulations and end-use checks instead). Confidence: low. The safe product behavior is a flag when an attachment or line says "ITAR" or "export controlled": do not forward the files outside the quoting team, and ask the buyer about handling.

PMI (positive material identification). PMI checks the metal's actual chemistry with a handheld analyzer, not the paperwork. It is governed by API RP 578 and ASTM E1476; full (100%) checks are typical for critical alloy systems, sour service, and high-temperature piping, with sampling elsewhere (Mistras, Fabrico, vendor pages, medium-low; I did not read API 578). If a line says "PMI," the open question is whether RudCo performs it, supplies a report, or whether the buyer does it on receipt.

NACE MR0175 / ISO 15156 (sour service). For carbon and low-alloy steel the common limit is 22 HRC maximum hardness, which the buyer will usually want shown on the MTR.

- Several sources give 22 HRC (about 248 HV10 or 237 HBW); one forum quoting the standard says it applies to carbon and low-alloy steels under 1% nickel in normalized, normalized and tempered, or quenched and tempered condition (Project Materials, TWI; medium).
- The standard itself does not require a conformance statement on the certificate, according to a NACE interpretation quoted in search results (I could not trace the page; low). So a purchase order should ask for hardness values and heat-treat condition, not just "NACE compliant." Austenitic stainless base metal has no HRC limit per the sources I found, but weld and cold work rules apply.

Costliest reading errors

Section 4

My ranking, by whether the mistake can be fixed after the metal ships. This is judgment from the consequences above. I found no public study that prices misread RFQ lines, and I did not find one gauge-error incident with a published dollar cost.

1. **Wrong or missing origin (DFARS, Buy America).** Metal melted abroad cannot be repaired. The buyer may have to reject it, and a supplier that certifies wrong has legal exposure. Related enforcement example, on the *testing* side: Bradken paid $10,896,924 to resolve False Claims Act allegations that an employee falsified test results on steel components for Navy vessels (DOJ press release, US Attorney, W.D. Wash.; fetched via search summary). That case is about falsified data, not misreading an RFQ. It shows how seriously certs are treated.
2. **Wrong grade or wrong spec family.** 304 vs 316, A36 vs A572-50, A500 vs A513, A53 vs A500. Paperwork cannot turn one into the other. The cleanest public stake example is a materials failure from chemistry alone: the 2012 Chevron Richmond refinery fire traced to sulfidation corrosion in a carbon steel pipe with low silicon (CSB, Process Engineering). That was an inspection and design failure, not an RFQ reading error. I include it only to show that chemistry is not cosmetic.
3. **Missing certification the buyer needed.** Often fixable if the heat was tracked and the mill cert exists; not fixable if cut stock lost its heat number. Depends on RudCo's traceability, which I could not see.
4. **Wrong gauge table.** Silent, because the quote looks normal. 4.5% thick on stainless, up to 24% thin on aluminum read with the steel table. Whether it fails depends on the buyer's tolerance.

5. **Wrong unit or position (x4 as quantity vs dimension; feet vs inches; 4x8 ft vs 4x8 in).** Easy for a rule to catch, and large when missed, so a parser should never guess.
6. **Sheet vs plate at the boundary (3/16 in).** Changes price, flatness, and the spec; usually caught on the first email back.

What this means for the two decisions

Section 5

- **Put first: reading accuracy with the table shown, then flags.** The gauge table, the unit rules, and the one-way substitution rules are deterministic and checkable. Judges "test it against real cases"; these are the cases they can test.
- **"Catches hidden requirements" can be a strong angle only if it is precise.** Evidence for: the requirements are real, costly, and absent from tools that only parse lines (the existing-tools track covers that). Evidence against: many triggers sit outside the line, the one real RFQ I found shows none, and a tool that flags everything breaks the PRD's own test ("No clear line gets a question it did not need"). Frame each flag as "this implies X because of Y; confirm," never as a legal ruling.
- **A cheaper way to show the same strength:** a flag with its trigger text highlighted (for example "domestic" or `SA-240` or the ship-to) combines the hidden-requirements angle with "the quote shows its work."
- **Fix in the PRD:** the example pair for "the same item written five ways" lists two items that differ by 4.2% in thickness. Use the test set below instead.

Test set

Section 6

How to read this. `V` = verbatim text from a cited public document (the line text is exact; I removed nothing). `C` = constructed by me from cited rules; the text is mine, the facts it tests are cited. Each line gives the standard reading and, when the reading is genuinely open, the question to ask. All thicknesses are inches. Gauge values come from the Metal Supermarkets chart unless stated. Unit conversions and gaps come from domain_rules_gauge_calc.py.

Sources for verbatim lines: `Iowa` = State of Iowa RFQ 245, issued 2018-08-28. Its lines (numbered 1 to 49) name no mill test report, certification, or metal origin requirement. The only origin wording is a general preference for Iowa products when bids tie. `SacRT` = Sacramento Regional Transit District steel road plates solicitation. `UAlberta` = catalog listing.

Verbatim lines

T1. V, Iowa line 20, quantity 19 EA `Stainless Steel Sheets Type 304-2B` / `Stainless Sheets 4 x 10` / `48" x 120" long x 16 Gauge`

- Reading: stainless steel, grade 304, finish 2B, sheet, thickness **0.0625** (stainless table; some suppliers list 0.063 or a different decimal, see section 1.1), width 48, length 120 (10 ft), quantity 19, spec not written (A240 is the usual spec family).
- Ask: "Is 16 gauge 0.0625 in nominal, or do you need a specific decimal? Do you accept 304/304L dual cert? Do you need a mill cert?"
- Why it is here: a clean reading with a gauge-table lookup and a quiet, optional cert question.

T2. V, Iowa line 9, quantity 2 EA `Cold Rolled Sheets QC/ A366` / `48" x 120" x 14 Gauge`

- Reading: carbon (mild) steel, cold-rolled sheet, thickness **0.0747** (mild steel table), width 48, length 120, quantity 2. "QC" probably means a quality designation; A366 is an older cold-rolled sheet spec whose current status I did not verify.
- Ask: "Is 'QC / A366' commercial quality? Current specs for cold-rolled sheet are A1008-family; do you accept a current equivalent?" (The A1008 and A366 facts are general knowledge, not checked here; a human should confirm before the demo asserts them.)

T3. V, Iowa line 1, quantity 24 EA `Square Tube A513-A500 Grade B` / `2" x 2" x 24' long x .188 thick`

- Reading: carbon steel, square tube, 2 x 2 in, wall **0.188**, length 288 in (24 ft), quantity 24. Spec is **ambiguous**: A513 (mechanical, no property requirements) and A500 (structural, verified properties, ships with MTR) are written together.
- Ask: "Structural tube (A500 Grade B) or mechanical tube (A513)? Do you need the mill test report?"

T4. V, Iowa line 11, quantity 10 EA `Expanded Stainless 4 x 8` / `Expanded Stainless Flat` / `48" x 96" x 13 GA x 1 1/2" Diamond`

- Reading: expanded stainless steel metal, flat, 48 x 96 in (4 x 8 ft), "13 GA", 1 1/2 in diamond, quantity 10. Grade not stated.
- Cannot resolve: 13 gauge does not appear in the stainless sheet table I cited, and expanded metal sells by its own stock numbers. **Thickness: unknown.**
- Ask: "What is the strand thickness in inches, or the expanded-metal stock number? What stainless grade?" This is a case where the right behavior is to say "I do not know," not to look something up.

T5. V, Iowa line 6, quantity 10 EA `Hot Rolled Strips CQ-A36` / `3" x 20' long x 1/8" thick`

- Reading: carbon steel, hot-rolled flat, width 3, thickness **0.125**, length 240 in (20 ft), quantity 10.
- Ask: "Hot-rolled A36 flat bar, or commercial-quality strip? 'CQ' and 'A36' are different specs." Form is flat bar or strip (0.125 is under the 0.230 in A6 bar line quoted in section 1.5).

T6. V, Iowa line 13, quantity 10 EA `Stainless Tubes Type 304` / `1 1/4" x 1 1/4" x 20' long x .120 thick`

- Reading: stainless 304, square tube, 1.25 x 1.25 in, wall **0.120**, length 240 in, quantity 10. No spec written.
- Ask: "Welded or seamless? Mechanical tubing or pressure? Finish (mill or polished)?" Note .120 is a decimal wall, not a gauge.

T7. V, Iowa line 23, quantity 5 EA `Aluminum Flat 6061-T6` / `1" x 12' Long x 1/8" thick`

- Reading: aluminum 6061, temper T6, flat bar, width 1, thickness **0.125**, length 144 in, quantity 5.
- Ask: "Extruded flat bar, or rolled/cut from sheet?" (B221 covers extrusions; B209 covers flat-rolled sheet and plate.) Low-cost question.

T8. V, SacRT road plates, quantity two `ASTM A36 or A572 Gr. 50 equivalent hot-rolled carbon steel` / `8 feet x 20 feet x 1 inch thick` / `Corrosion-resistant coating (primer plus industrial enamel, galvanized, or equivalent)`

- Reading: carbon steel plate, hot-rolled, thickness **1.000**, width 96 in, length 240 in, quantity 2. Grade: either A36 or A572-50 is acceptable ("or"), so a dual-cert plate fits.
- Ask: "Which coating: primer and enamel, or galvanized?" The "equivalent" wording invites a ruling on what counts as equivalent; the safe answer is dual-certified A36/A572-50.
- Hidden requirements in the listing: none stated. The full PlanetBids document may add some; I did not see it.

T9. V, UAlberta catalog listing (not an RFQ; one item written three ways) `Stainless Steel Sheet 304` / thickness `1/16 in` / `0.063 (16 Ga.)` / `2B Mill Finish` / `12 in x 12 in`

- Reading: 304, sheet, 2B, thickness **0.0625** (1/16 in; also the stainless 16 gauge value), 12 x 12 in, quantity 1. The listing's 0.063 rounds 0.0625 by 0.8%.
- Ask: none. This is the positive control: three notations, one thickness.

Constructed lines

T10. C, from the PRD's example, quantity missing `304 SS sheet 16ga 4x8`

- Reading: stainless 304, sheet, **0.0625**, 48 x 96 in, spec not written, finish not written, quantity missing.
- Ask: "How many sheets? Finish: 2B, No. 4, or other? Do you accept 304/304L dual cert?"

T11. C, the PRD's second example `Stainless 304, 0.060 in, 48 x 96`

- Reading: stainless 304, **0.060** exact as written, 48 x 96 in. Under the stainless table, 16 gauge is 0.0625 (4.2% thicker). 0.060 matches 16 gauge *mild steel* (0.0598).
- Ask: "Do you need 0.060 in exactly, or is 16 gauge (nominal 0.0625) acceptable?" This is the gauge trap in one line.

T12. C, material missing `16 ga sheet 4x8 qty 20`

- Reading: **cannot be read.** Four tables apply: mild steel 0.0598, galvanized 0.0635, stainless 0.0625, aluminum 0.05082. Width 48, length 96, quantity 20.
- Ask: "What material: carbon steel, galvanized, stainless (which grade), or aluminum (which alloy)?"

T13. C, galvanized gauge `16 ga galv 4x10 x 25`

- Reading: galvanized steel, sheet, **0.0635** (galvanized table; 6.2% thicker than mild 16 gauge), 48 x 120 in, quantity 25.
- Ask: "Which coating weight (for example G60 or G90)?" (Coating designations are general knowledge; not sourced here.)

T14. C, aluminum gauge and temper `14 ga 5052 alum 4x8 x10`

- Reading: aluminum 5052, sheet, 48 x 96, quantity 10, thickness **0.06408** (Metal Supermarkets) or **0.063** (Wikipedia). Spec likely B209.
- Ask: "Which temper (H32, H34, other)? Do you need 0.063 in or 0.064 in, or will either do?" The 1.7% gap between published tables is the reason to ask.

T15. C, quantity vs dimension `3/16 304 plt 4x8 x4`

- Reading: stainless 304, buyer says plate, thickness **0.1875**, 48 x 96 in, and a trailing `x4` that could be quantity 4 or part of a size.
- Ask: "Is 4 the quantity? Hot-rolled plate or cold-rolled 2B sheet at 3/16?" 3/16 sits between the 1/4 in and 3 mm sheet/plate lines. Here the buyer said "plt", so form is plate; confirm finish.

T16. C, contradiction between grade and strength `1/2 A36 PL 5x10 x3 min yield 50 ksi`

- Reading: carbon steel plate, thickness **0.500**, 60 x 120 in, quantity 3, grade A36. A36 guarantees 36 ksi minimum yield.
- Ask: "Do you need A572 Grade 50, or A36/A572-50 dual-certified plate?" Source: AISC via search. A hidden requirement: the stated strength implies a spec change.

T17. C, ASME code material with a 3.1 cert `SA-240 316L 3/8 plt 60x120 x2 w/ 3.1 cert`

- Reading: stainless 316L, plate, thickness **0.375**, 60 x 120 in, quantity 2, ASME code material SA-240, EN 10204 3.1 certificate.
- Ask: "Must the cert show the SA-240 designation (not only A240), and the 316/316L dual cert? Do you need a code stamp or PMI?" Hidden requirements flagged: ASME SA (Wermac), 3.1 (Material Welding).

T18. C, pipe with PMI `2in sch40 304/304L A312 smls 20ft x30 PMI`

- Reading: stainless 304/304L dual, seamless pipe, nominal size 2 in, schedule 40, length 240 in, quantity 30 (lengths? feet?), spec A312. Wall thickness comes from the schedule table, which is not in my sources, so I do not state it.
- Ask: "Is 30 the number of 20 ft lengths, or total feet? Do you need the PMI report with the shipment, or only performed?" Hidden requirement stated: PMI.

T19. C, contradiction between shape and spec `2x2 sq tube A53 Gr B .120 24ft`

- Reading: **cannot be read as written.** A53 is round pipe only (Steel Tube Institute); square tube is A500 or A513.
- Ask: "Do you need square tube (A500 structural or A513 mechanical), or round A53 pipe?"

T20. C, DFARS stated `304 plate 1/4 48x96 A240 DFARS compliant melted USA 3.1 x3`

- Reading: stainless 304, plate, thickness **0.250**, 48 x 96 in, quantity 3, spec A240, EN 10204 3.1, domestic melt.
- Hidden requirements stated outright: DFARS 252.225-7009 (acquisition.gov), melt origin on the MTR.
- Ask: "Is a qualifying-country melt acceptable, or US only?" (The clause allows the US, outlying areas, or a qualifying country.) And an internal check: does the stock on hand have a heat whose cert shows melt country?

T21. C, DFARS implied by context Context: ship-to is a defense contractor; PO carries a "DO-A1" priority rating. Line: `304 rd bar 2in A276 12ft x10`

- Reading: stainless 304, round bar, 2.000 in diameter, length 144 in, quantity 10, spec A276.

- Hidden requirements implied: 304 is a specialty metal by chemistry (chromium well above 0.25%), a defense ship-to and rating suggest the DFARS clause flows down, and a DPAS rating carries an acceptance deadline.
- Ask: "Does the PO include DFARS 252.225-7009 or a melt-origin requirement?" This tests the context-trigger path with no keyword in the line.

T22. C, Buy America with a coating step `L3x3x1/4 x 20' A36 galv x40 Buy America state DOT bridge`

- Reading: carbon steel angle, 3 x 3 x 0.250 in, length 240 in, quantity 40, A36, galvanized.
- Hidden requirement stated: Buy America. The rule covers the galvanizing step, and the MTR must show melt and manufacture location (23 CFR 635.410, Caltrans).
- Ask: "Where is the galvanizing done, and does your agency require specific certification wording?"

T23. C, sour service `1/2 A516-70 plate 72x120 NACE MR0175 HIC tested x2`

- Reading: carbon steel plate, thickness **0.500**, 72 x 120 in, quantity 2, spec A516 Grade 70 (pressure vessel plate; general knowledge, not sourced here), NACE MR0175, HIC tested.
- Hidden requirements: hardness limit (22 HRC max for carbon steel), heat-treat condition, HIC test results. The NACE standard does not itself require a certificate statement, so the buyer should say what the MTR must show.
- Ask: "Do you need hardness values and heat-treat condition on the MTR, and a separate HIC test report?"

T24. C, export-controlled attachment Line: `6061-T651 plate 2 x 12 x 24 x4`, with an attached drawing marked "ITAR controlled technical data."

- Reading: aluminum 6061, temper T651, plate, thickness **2.000**, 12 x 24 in, quantity 4.
- Hidden requirement: the attachment is export-controlled data. Confidence is low (see section 3). Flag, do not interpret.
- Ask: "Please confirm handling rules (US persons only, domestic suppliers only) before we circulate the drawing."

T25. C, a clear line with a gauge on tube (negative control) `1-1/2 sq tube A500 C 11ga 24ft x8`

- Reading: carbon steel, square tube, 1.5 x 1.5 in, wall **0.120** (tube wall table, 11 gauge), length 288 in, quantity 8, spec A500 Grade C. A500 tube ships with an MTR (Atlas Tube), so the flag is informational.
- Ask: **none.** The right behavior is to read it, show the table used, and not ask.

Gaps and source quality

Section 7

- **Paid or blocked sources:** ASTM, ANSI, SAE, and API documents were not readable. ASTM store pages and AISC PDFs returned 403 to my fetch tool. Spec claims rest on distributors, trade groups, forums, and search-result summaries of pages I could not open. Where a claim rests only on a search summary, the text says so.
- **Not verified in this session** (general industry knowledge, flagged in the lines that use them): A312 scope, A366 status and its modern equivalent, A516-70 as pressure vessel plate, G60/G90 coating designations, B221 scope, AMS specs, the exact A6 plate definition.
- **Not found:** a public dollar figure for a misread RFQ; a public statement on whether mill-form titanium or stainless is USML-controlled; a standards-body gauge chart (distributor and Wikipedia charts agree with each other).
- **One RFQ source for messy lines.** The Iowa document is a real, public RFQ from 2018 (line numbers 1 to 49), written by a state buyer and not by a plant engineer. It is cleaner than most email RFQs. Anything worse (PDF scans, abbreviations, mixed units) would come from the voice-of-user tracks.
- **Reddit:** not fetched, per instructions.

Cleanup needed

Section 8

None. Files I created in the research folder: `research_domain_rules.md` and `domain_rules_gauge_calc.py` (the script behind the percentages and conversions). A scratch copy of the script sits in the session scratchpad and needs no action.

Appendix 5: Existing Tools and Price Benchmarks

Needs action and bottom line

Track	existing tools (question 5) and price benchmarks (question 6) from the research brief.
Written	2026-10-10. Every vendor number below is a vendor claim unless a third party is named.

Two of the four angles look open; one is partly taken. Across about 15 tools, no vendor page I could read says it shows where each field came from in the buyer's words, sends the buyer a clarifying question, or flags certifications the spec implies. One vendor says it learns customer nicknames.

"Learns the buyer's shorthand" is the most contested angle. Proton.ai says its system "learns your catalog, your customers' nicknames, and your reps' editing patterns over time." That is a claim on a general distribution tool, not a metals tool, and I found no user evidence either way.

"Catches hidden requirements" looks the emptiest. No tool I read mentions mill test reports, DFARS, Buy America, or spec-implied certifications. The security and compliance badges on vendor pages (SOC 2, ITAR hosting, CMMC) describe the vendor's own software, not the buyer's requirements.

"Shows its work" and "one-click buyer questions" are open on the pages I read, but thinly. A blog post describes the pattern (Stacksync), and one vendor lists a Message Center to centralize communication as a planned update (DigiFabster). Neither ships it as far as the pages say.

Big caveat. I read marketing pages, YC launch posts, and trade press. A feature missing from a page may still exist in the product. No tool was tested. Treat "open" as "not claimed publicly," not "does not exist."

For the price explanation, claim less than "cheapest." Public sources give the metal's market cost, not competitors' delivered prices. The honest sen-

tence is "this price tracks the published base-metal cost plus RudCo's margin." See Part B.

Contents

Part A: Existing tools

Part B: Public price sources

Cleanup needed

Part A: Existing tools

Section 1

What I searched

- AI RFQ and quote automation for metals service centers, distributors, and fabricators (about 14 searches).
- YC and venture-backed companies in industrial distribution quoting.
- Metals ERP vendors (service center software).
- User complaints, on G2, Capterra, and trade press.
- Per-field source tracing in RFQ tools.
- Reddit was not searched or fetched, per the rules.

The comparison table

Legend: **Yes (claim)** means the vendor says it. **Not stated** means I read the page and found nothing. **Planned** means the vendor says it is coming.

Tool	What it does	Who it serves	Field-level provenance	Asks the buyer clarifying questions	Learns per-customer shorthand	Flags compliance or cert needs
Avent (YC S25)	Watches email, fax, and phone; builds quotes from past pricing; matches POs to quotes; writes into the ERP	Industrial distributors and manufacturers, North America and UK	Not stated	Not stated	Partly: builds quotes from "past pricing and tribal knowledge" (claim)	Not stated
Hexa (YC, 2026)	Reads RFQs by email, PDF, Excel, fax; matches SKUs and pricing; drafts quotes for rep approval	Mid-market industrial distributors with legacy ERPs	Not stated	Not stated	Not stated	Not stated
Mercura (YC W25)	Extracts product details from RFQs and specs; matches to the catalog using the customer's sales history; enters the quote in the ERP	HVAC, electrical, and plumbing distributors (per its YC page); tender and BOQ work per a profile	Not stated	Not stated	Partly: matching uses sales history (claim)	Not stated
Proton.ai	Turns email, PDF, Excel, handwritten notes, and competitor part numbers into draft quotes or orders; rep approves	Wholesale distributors	Not shown; says it flags low-confidence matches and missing fields	Flags missing info for the rep; does not say it asks the customer	**Yes (claim):** learns "your customers' nicknames"	Not stated; lists SOC 2 for itself

Tool	What it does	Who it serves	Field-level provenance	Asks the buyer clarifying questions	Learns per-customer shorthand	Flags compliance or cert needs
CloudForge	AI sales and procurement platform for metals: outbound sales, CRM, vendor management. Funded $3.95M, Feb 2026 per a coverage summary	Metal service centers, fabricators, machine shops	Not stated	Not stated	Not stated	Not stated
Nox Metals	A metal service center that built its own instant online quoting and "AI-parsed material lists" (an investor's write-up)	Its own customers	Not stated	Not stated	Not stated	Not stated
Metal Networks.AI (JAQi)	Paste email inquiry text; the engine "deciphers" it and searches a catalog or inventory	Metals buyers and distributors; company status is unclear (sources conflict)	Not stated	Not stated	Not stated	Not stated
Frankstahl RFQ processor (Wonderful)	German steel seller's agent: parses free-form RFQ emails, classifies lines into about 350 product types, configures variants, posts a priced quote to the CRM. Prefers rules to the model "wherever possible" (per a search summary of the docs; the page needs a login)	One steel seller	Not stated	Not stated	Not stated	Not stated
Invera INVEX "Ivy"	June 2026 AI layer on a metals ERP: help, natural-language questions on ERP data, "agents"	Metal service centers on INVEX	Not stated	Not stated	Not stated	Not stated
Paperless Parts	Quoting and estimating; added an AI step that reads forwarded RFQ emails and builds line items (The Fabricator, via search)	Job shops, fabricators, and some metal service providers	Not stated	Not stated	Not stated	Not stated; "Security and Compliance" is

Tool	What it does	Who it serves	Field-level provenance	Asks the buyer clarifying questions	Learns per-customer shorthand	Flags compliance or cert needs
						about the vendor
DigiFabster AI Quote Agent	Reads RFQ inbox, drafts a quote for review. Beta as of Dec 2024	Machine shops and metal fabricators	Not stated	**Planned:** a "Message Center"	Not stated	Not stated; ITAR hosting is about the vendor
SpeedQuoteAI	Email monitoring, auto quote "in under 2 minutes" (claim)	CNC and sheet metal fabricators	Not stated	Not stated	Not stated	Not stated
AIQuote (LogisticsMRO)	Parses RFQs, checks inventory, prices, auto-sends above a confidence threshold, else routes to sales	Aviation parts distributors	Not stated	Not stated	Not stated	Not addressed in the article
Cassidy RFQ agents	Extracts fields from emailed PDFs, BOMs, and Excel; confidence scores for human review	Electronics and pump makers (templates)	Confidence scores only; no source links stated	Not stated	Not stated	Not stated
Metals ERPs: 4GL Steel Manager III, Crowe Metals ERP, INVEX, Metalogic	Inventory, quoting, and order modules for service centers. Quoting is a screen in the ERP; RFQ email intake is not described	Metal service centers	Not stated	Not stated	Not stated	Not stated

What the table says

- **The basic job is already being sold to general distributors.** Avent, Hexa, Mercura, and Proton all say they turn messy RFQs into draft quotes with a rep approving. The RFQ-to-draft-quote loop is not an unclaimed idea. Our demo cannot win on "AI reads the RFQ and drafts a quote."
- **Almost nobody is metals-specific at the line level.** The metals-named players (CloudForge, Nox, Invera, Metal Networks.AI) do sales, online ordering, ERP help, or catalog search. Frankstahl is the closest to our case, and it is one German seller. I found no public product that handles gauge, grade, and ASTM reading for a US metals supplier.
- **Provenance is rarely named as a feature.** The nearest items are confidence scores (Cassidy), a note that a low-confidence match gets flagged (Proton), and a design pattern in a blog post that keeps "source references" so each extracted detail traces to its document (Stacksync). A Softomate FAQ says it shows candidate matches next to the source description when a request is unclear. I found no vendor that links each field to the highlighted words in the RFQ.
- **Buyer-facing questions are unclaimed.** Proton and the Stacksync pattern send open questions to an internal reviewer. DigiFabster lists a Message Center (to centralize communication; the article does not say who uses it) as planned. None says it drafts a multiple-choice question to the buyer.
- **Trust is a stated worry in the sources.** A Forgepoint listing's critics flag that wrong AI quotes cause margin leakage and disputes (per a search summary). An Infosys BPM piece argues quoting flows without human review create accountability gaps and compliance exposure. The DreamzTech case study notes that grade, thickness, unit of measure, and delivery location change the right answer.

Vendor claims, kept separate

All of these are claims. None is independently verified.

- Avent: "over $200,000 worth of quotes every day" (YC launch); 50% less admin work, per the founder in HRAI.
- Hexa: revenue growth "over 10%," response "from hours to minutes," "live in days" (YC launch). A YC tier-list review found no screenshots, demo, or named customers.
- Proton: quotes from "15 minutes to under 60 seconds," "85% first-time match rate on cross-referenced parts" (Proton).

- Mercura: "process RFQs in minutes instead of days" (YC launch); 60+ customers and a $2.1M seed come from secondary profiles.
- AIQuote: one unnamed distributor gained "over $630,000" in four months; the article's URL says $350,000 (article).
- Paperless Parts: a customer testimonial says quoting time cut by 80% (Paperless Parts).

User complaints and gaps

- **Avent has no third-party reviews.** Its G2 profile showed 0 reviews in a search summary.
- **Hexa has no independent user reviews** that I could find (search).
- **Paperless Parts reviews are about the quote builder, not the AI.** G2 and Capterra complaints include slow lookup, limited data access for sales analysis, and a quote builder that expands one item at a time (SaaSworthy summary). I found no review that tests AI extraction accuracy.
- **Metals ERP vendors: no user complaints found.** I searched for Capterra and G2 complaints on Steel Manager, INVEX, and Metalogic and got only company profiles.
- **Gap to state plainly:** I found no documented complaint that an AI quote tool misread shorthand or missed a cert. The pain evidence for those belongs to the voice-of-user tracks, not this one.

What I could not confirm

- **Frankstahl's docs page needs a login.** I relied on a search summary of it.
- **Mercura's YC page names HVAC, electrical, and plumbing.** A profile site says it also does tenders. I treated the YC page as the primary source.
- **Metal Networks.AI's status is unclear.** PitchBook lists it as acquired or merged, while Craft lists it as active.
- **No tool was tested.** Several vendor sites returned 403 to my fetches (Proton's MDM write-up, G2 seller pages).

Part B: Public price sources

Section 2

This feeds Yuval's catalog and price work. Status checked 2026-10-10 where I could open the page.

Summary table

Source	Covers	Update frequency	Free and reachable?	How a quote explanation can cite it
FRED / BLS PPI, steel mill products (WPU1017)	US producer price index for steel mill products, 1982=100. Latest shown: 381.162 for August 2026	Monthly; next release Oct 15, 2026	Free, no login for the page. FRED asks for citation	"Steel mill product prices were at index 381.2 in Aug 2026 (BLS PPI via FRED)"
FRED / BLS PPI, aluminum mill shapes (WPU102501)	US producer price index for aluminum mill shapes. Latest shown: 370.902 for August 2026	Monthly; next release Oct 15, 2026	Free	Same pattern. There are finer series such as aluminum sheet and strip
IMF global nickel price on FRED (PNICKUSDM)	Monthly average nickel, USD per metric ton. Latest shown: 16,632.36 for July 2026	Monthly, about 6 weeks lag	Free; IMF copyright, citation required	"Nickel averaged $16,632 per tonne in July 2026 (IMF via FRED)" for a stainless surcharge story
IMF global aluminum price on FRED (PALUMUSDM)	Monthly average aluminum, USD per metric ton. Latest shown: 3,158.27 for July 2026	Monthly	Free; citation required	Same pattern
LME reports and data	Official prices for aluminium, nickel, copper, zinc, and others	Daily, next-day delayed on the website	Free **with a registration** on LME.com, per a search summary of its pages; same-day data needs a paid licence (a delayed tier was listed at USD 210 a month)	"LME official nickel cash price on [date], next-day delayed." Check the LME terms before reusing numbers in a public demo
CME Midwest HRC steel futures (CRU index)	Cash-settled against the CRU US Midwest hot-rolled coil index	Daily settlements	CME pages show quotes delayed at least 10 minutes. A Fastmarkets help page says CME has a free 10-minute delay (search summary). I could not open CME's pages to confirm	"Front-month HRC futures settled at $X on [date] (CME)." CME says web data is reference only

Source	Covers	Update frequency	Free and reachable?	How a quote explanation can cite it
CME Aluminum Midwest premium futures	The US Midwest premium on top of the LME aluminum price, settled on a Platts assessment	Daily	Contract page is free. The Platts assessment behind it is paid. Delayed-quote terms not confirmed	"Aluminum price = LME base plus Midwest premium of about X cents per lb"
Nucor Consumer Spot Price	Nucor's weekly hot-rolled coil spot price, issued Mondays since April 2024	Weekly	I found no public nucor.com price page. Nucor says sheet price extras are in its NucorNow portal. Weekly values are reported by SteelOrbis	"Nucor's posted HRC spot was $X per ton on [date]" if Yuval can find the live number
North American Stainless news page	Monthly fuel surcharge notice (58% for flat and long products effective Oct 1, 2026) and base price announcements	Monthly, posted near month end	Free, no login	Cite as a surcharge line on stainless quotes: "NAS fuel surcharge 58% for October 2026." This is a fuel surcharge. I found no alloy surcharge table on the public page
Outokumpu alloy surcharges	Alloy surcharge by grade. The overview page says surcharges are "usually published month by month" and links an Americas page	Monthly	Overview is public. I could not open the Americas table, so I do not know if it is fully free	"Alloy surcharge for 304 is $X per lb for October (Outokumpu)" once Yuval confirms access
ATI flat rolled surcharge	Raw material surcharge announcements for stainless	Monthly	ATI posted notices as PDFs in the past. I found no current-month notice	Same as Outokumpu if reachable
MetalMiner Monthly Metals Index	Direction of price moves for carbon steel, stainless, aluminum, and others	Monthly	A search summary says free. I could not open this page (404), so confirm	"MetalMiner's Stainless MMI rose this month." Gives direction, not dollar prices

Source	Covers	Update frequency	Free and reachable?	How a quote explanation can cite it
USGS Mineral Commodity Summaries 2026	Annual supply, trade, and price notes for iron and steel, nickel, aluminum	Yearly, February	Free	Context only. Too coarse for a quote line
World Bank Pink Sheet	Monthly commodity prices including nickel and iron ore. I did not confirm aluminum in the editions I saw	Monthly	Free PDFs. I did not find the current edition	Backup for nickel if FRED is down
Paid: Fastmarkets, CRU, S&P Platts, Steel Market Update	The assessments most buyers and mills actually reference: steel, scrap, premiums	Daily to weekly	**Paid.** No public prices; "talk to us" sales pages	Name them as "the benchmark RudCo would license," not as data we show

How to use these in the demo

- **Best two for a free, live-looking demo:** FRED's BLS PPI series (steel mill products, aluminum mill shapes) and the IMF nickel and aluminum prices on FRED. They are free, open pages with named update dates, and they cite cleanly.
- **Best for stainless:** nickel drives the alloy surcharge, so the IMF nickel price plus a mill surcharge notice tells the story. NAS posts a public monthly notice, but it is a fuel surcharge, not the alloy surcharge. The alloy tables I found were third-party reports (Argus, Kallanish).
- **What the sources cannot say.** A PPI is an index with a 1982 base. It tells the rep which way the market moved, not what a competitor would quote on 16 gauge 304 sheet. The IMF prices are monthly averages of the raw metal. So the explanation line should read as market context for RudCo's price, for example "Nickel is X% higher than a year ago, so the 316 surcharge line rises." It should not read as proof RudCo is cheaper. That keeps the trust story honest.
- **Mark sample data on screen.** If the RudCo catalog and price list are made up, say so, per the product brief.

Gaps in Part B

- **LME's exact free terms.** The sources differ: one says next-day delay, another shows 15-minute delayed intraday quotes on price pages. Registration is needed for history. I could not open lme.com directly (403).
- **CME quote access.** I could not open the CME pages (timeouts). The 10-minute free delay comes from a Fastmarkets help page, not CME.
- **Nucor's live CSP.** The most recent figure I saw was April 2026, from SteelOrbis. Yuval needs to confirm where the current weekly number is posted.
- **Mill alloy surcharge tables.** I could not read current NAS, Outokumpu Americas, or ATI tables. A search result mentioned Outokumpu's October 2026 North American surcharges, but I did not verify the figures against Outokumpu's own page.
- **No arithmetic.** I did not compute year-over-year changes. If the demo shows a percentage, run it through a script on the series data.

Cleanup needed

Section 3

None. I created only this file and deleted nothing.